

2025秋季学期的《集成电路学科前沿》



清华大学 集成电路学院
School of Integrated Circuits, Tsinghua University

集成电路测试、可测性设计与芯片 生命周期健康管理

VLSI Testing, Design-for-Testability & Silicon Lifecycle Management

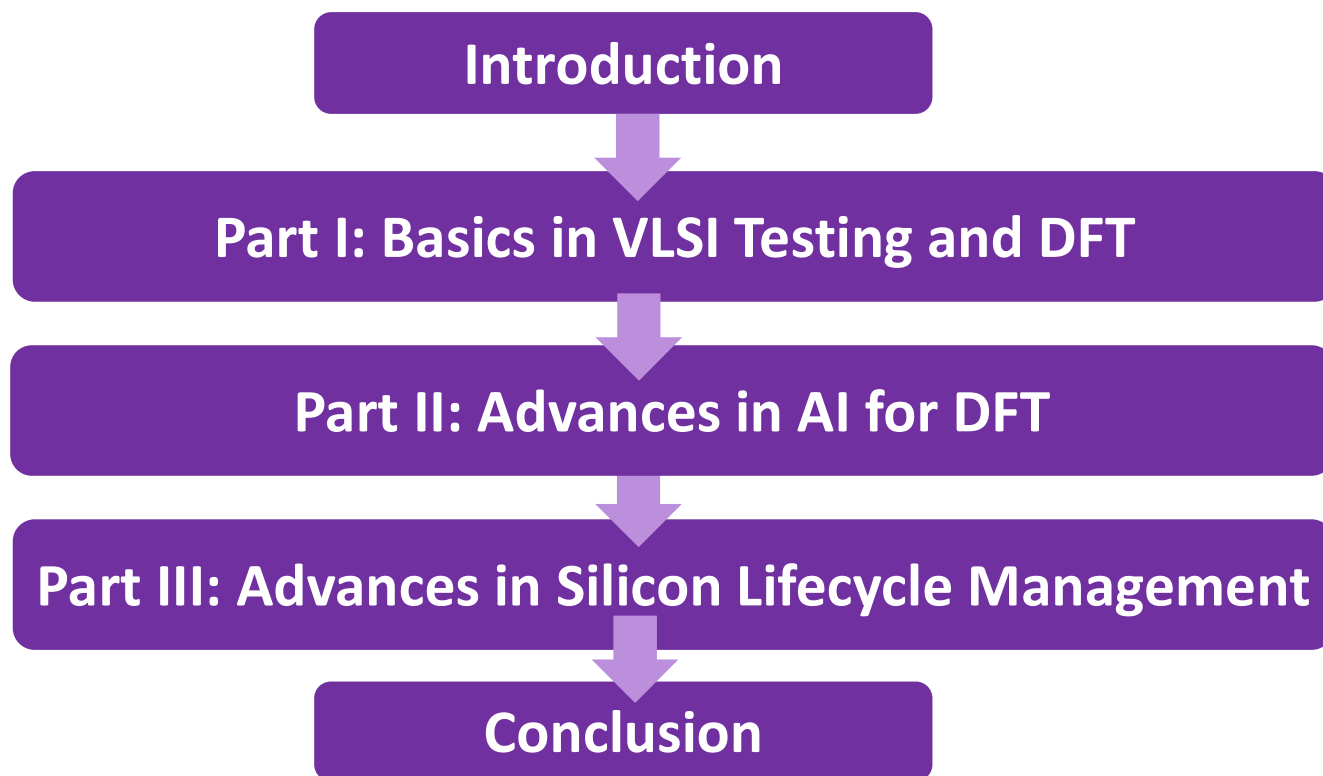
苏菲 Fei Su

Professor

School of Integrated Circuits
Tsinghua University

2025年12月

Lecture Content List



Lecturer



苏菲，原Intel芯片架构师/测试与可测性设计技术专家，拥有**20年**深耕半导体一线的研发经验，负责多代英特尔关键核心产品从130nm到<2nm的测试与可测性，涵盖从DFX架构/设计/验证到量产测试全过程。包括从第一代起的**Intel Atom®凌动处理器**系列，国际领先的**MIPI**移动产业处理器接口IP，第一代基于Intel **Foveros™ 3D集成**技术的芯粒Chiplet/SOC，第一代基于英特尔**18A制程**的CPU等。

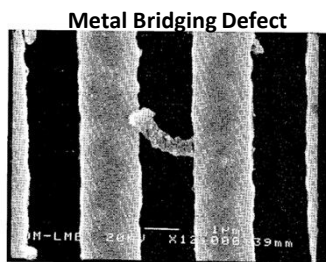
曾作为国际半导体研究协会（SRC）业界导师，指导多项产学研合作研究项目，并因突出贡献荣获2021年**SRC Mahboob Khan Outstanding Industry Liaison Award**。多次获得最佳学术论文奖，包括 IEEE 电路与系统协会 Outstanding Young Author Award、IEEE International Conference on VLSI Design Best Paper Award 等。技术创新获多项**国际专利**，涵盖**3D 芯片/芯粒（Chiplet）测试**、模拟芯片功能安全（Functional Safety）、智能传感器等领域。推动先进研究成果的标准化，参与多项 **IEEE 国际技术标准的制定**。

工作经历

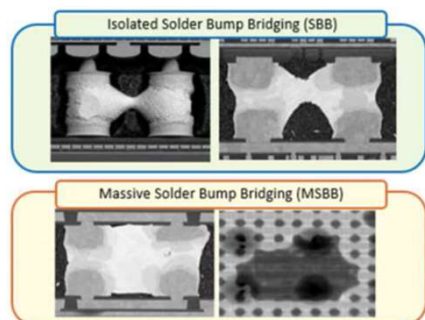
2025.8至今	清华大学集成电路学院，长聘教授、兴华讲席教授
2006.2 – 2025.7	美国INTEL英特尔公司，ARCHITECT 架构师
2023.11至今	美国亚利桑那州立大学，ADJUNCT FACULTY 兼职教授
2018.8 – 2021.8	美国杜克大学，TERM GRADUATE FACULTY 兼职教授

Part I: Basics in VLSI Testing & DFT

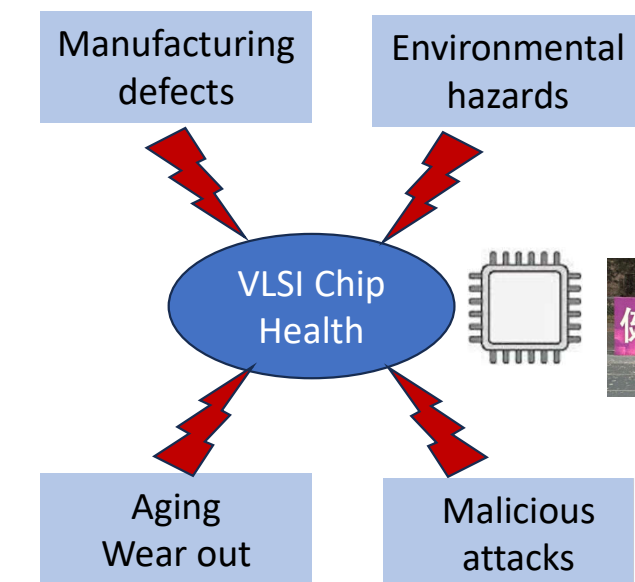
Testing: Why



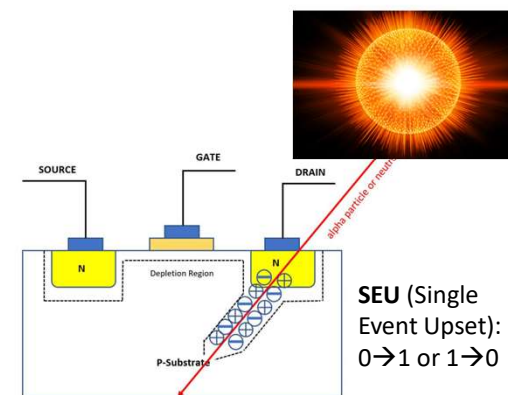
Source: IEEE IT **1992** Rodriguez-Montanes, R. et. al.



Source: IEEE VT **2023** Chakravarty, S.



Testing is key to ensure silicon health



Soft Error Caused by Radiation



Real-World Examples

Example #1: Hyper-scale data center SDE (Silent Data Error)/ SDC (Silent Data Corruption) issues

- Google and Facebook/Meta papers in 2021
- Caused by silicon health-related issues



Why Silent Data Errors Are So Hard To Find

Subtle IC defects in data center CPUs result in computation errors.

OCTOBER 11TH, 2022 - BY: ANNE MEIXNER

Example #2: Autonomous driving car recall

- “...China's market regulator indicated that Tesla will recall a total of 127,785 units of Model 3 cars in China due to **potential faults in semiconductor components** that might lead to collisions...”

Meta Silent Data Corruptions at Scale

After a few iterations, it became obvious that the computation of

$$\text{Int}(1.1^{53}) = 0$$

as an input to the `math.pow` function in Scala would always produce a result of 0 on Core 59 of the CPU. However, if the computation was attempted with a different input value set

$$\text{Int}(1.1^{52}) = 142$$

Google Cores that don't count

CEE: Corrupt Execution Error

For business reasons, we are unable to reveal exact CEE rates, but we observe on the order of a few mercurial cores per several thousand machines – similar to the rate reported by Facebook [8]. The problem is serious enough for us to have applied **many engineer-decades** to it.

Tesla to recall Model 3s in China due to faulty semiconductor components

Apr. 07, 2022 8:18 AM ET | Tesla, Inc. (TSLA) | By: SA News Team | 67 Comments

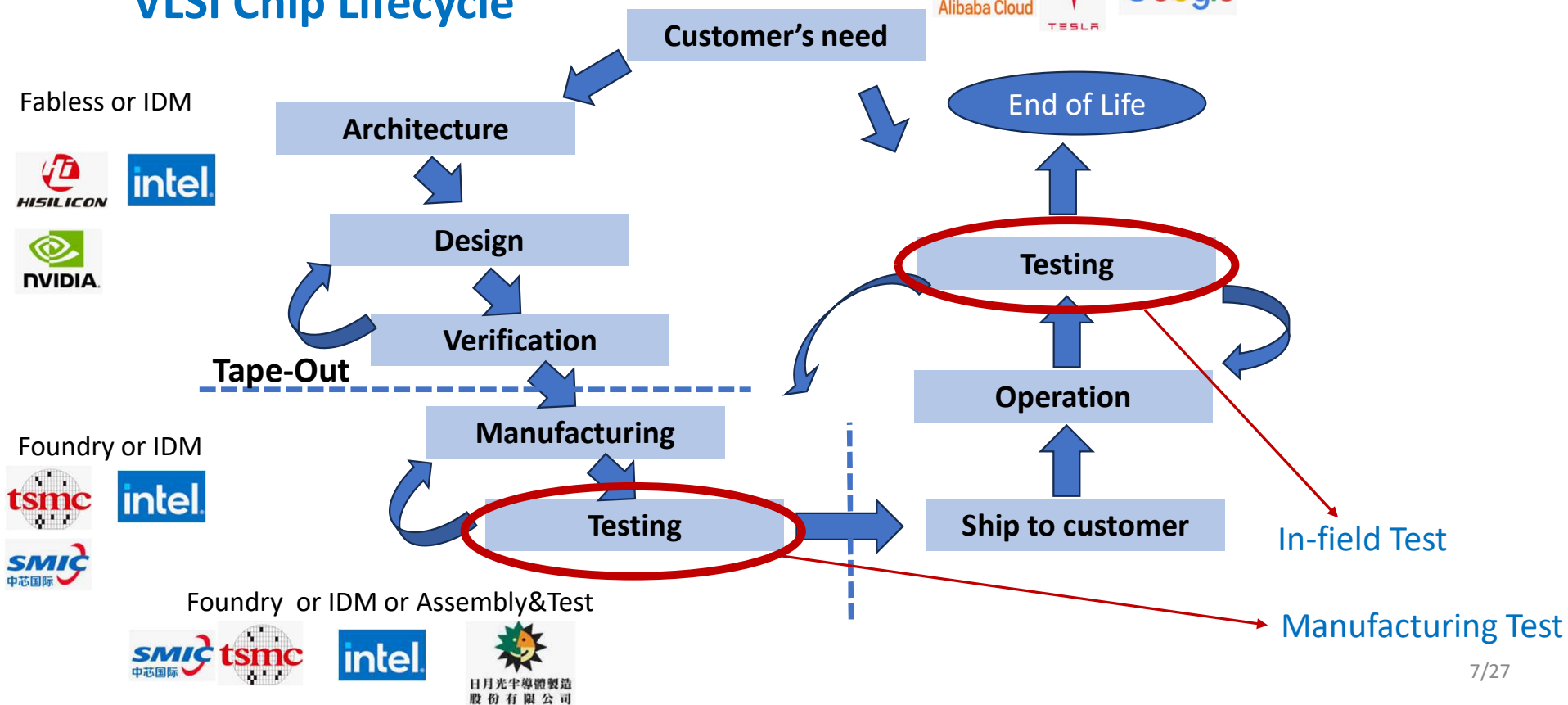


y_carfan/iStock Editorial via Getty Images



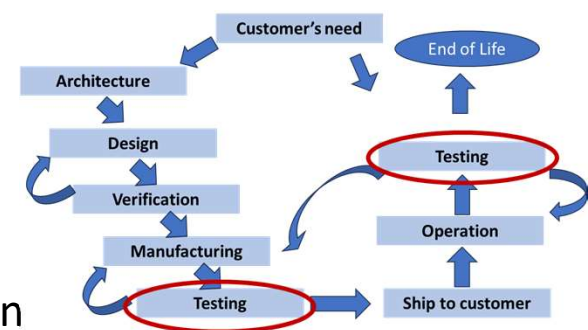
Testing: What

VLSI Chip Lifecycle



Testing: What

- **Manufacturing Test:** A manufacturing step that ensures that the physical device, manufactured from the design, has no *manufacturing defect*
- **In-Field Test:** A in-field step that ensures that the physical device, shipped to the customer, has no *service error* during the operation



Verify correctness of design

Target design bug

Performed prior to manufacturing

Performed by simulation, emulation or formal methods

Responsible for quality of design

Verify correctness of manufactured hardware

Target hardware defect

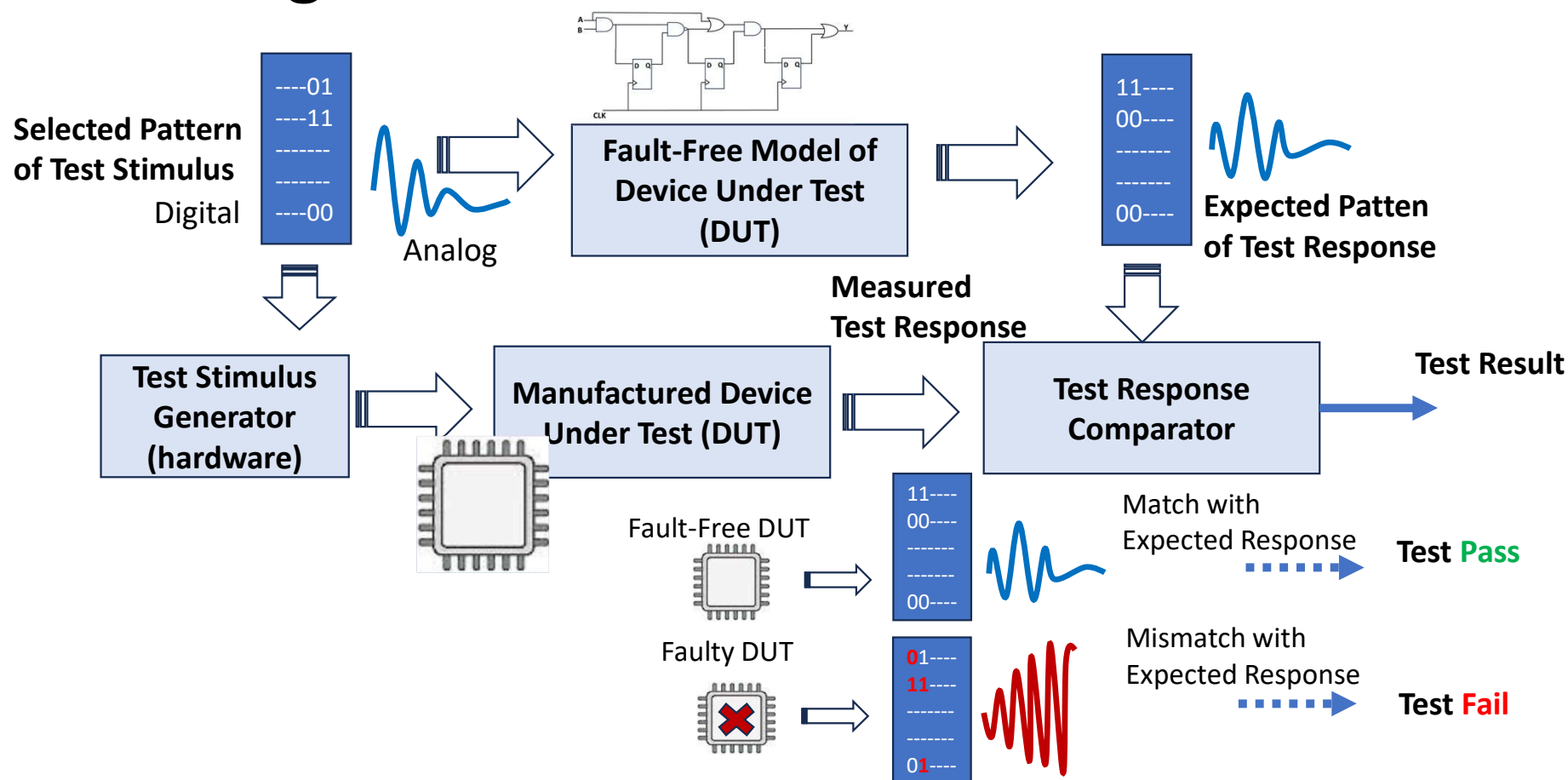
Test application performed on every manufactured device

Performed by testing methods (described later)

Responsible for quality of device

Testing: How

Testing Principle



Testing Approaches

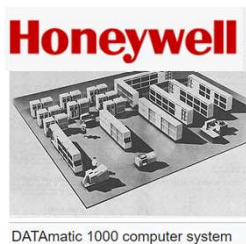
Testing Goal: to detect any defect/fault impacting DUT functionality → the appropriate test stimulus patterns?

- **Functional Test:**

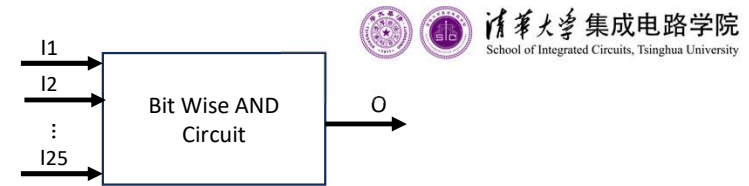
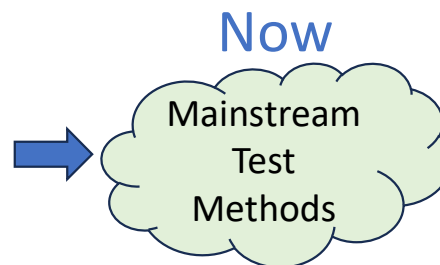
- Exercise the functions according to the DUT functional spec.

- **Structural Test:**

- No direct attempt to determine if the overall DUT functionality is correct
- Instead, to make sure that the circuit has been assembled correctly from some low-level building blocks (e.g., as specified in a structural netlist)
- Assume: the correct assembly of the DUT elements → DUT should function correctly

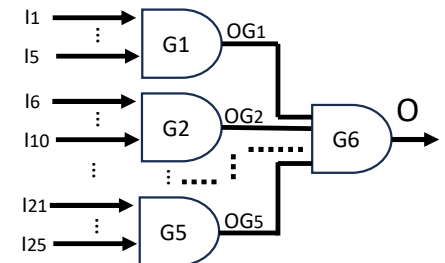


Eldred (1959) – First use of structural testing for the Honeywell Datamatic 1000 computer



Test Pattern #	Test Pattern (I25, ..., I2, I1)	Output (O)
1	000000000000000000000000	0
2	000000000000000000000001	0
...		0
2^{25}	111111111111111111111111	1

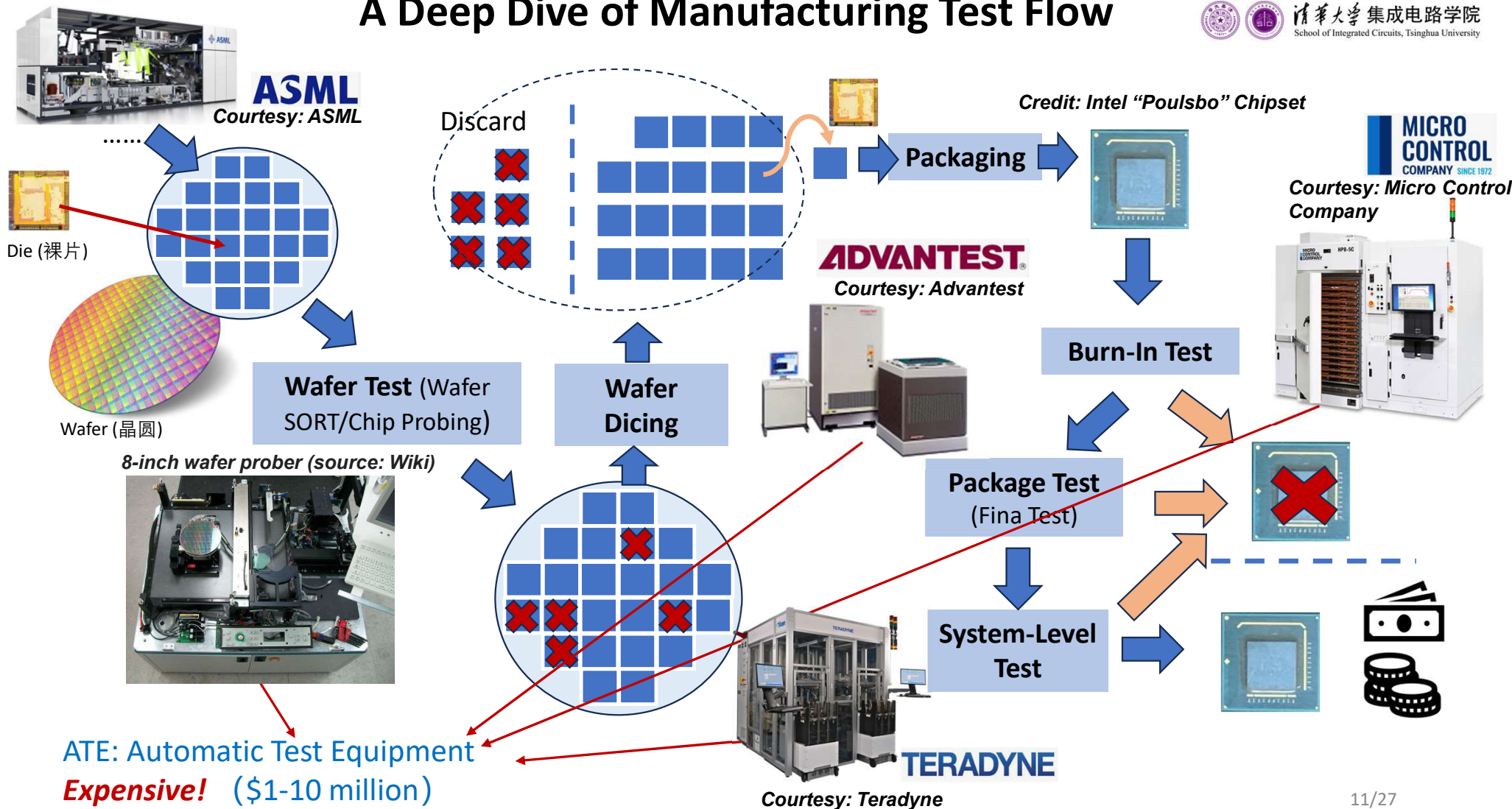
2^{25} Test patterns → 33 Seconds per chip (1 MHz tester: running 10^6 patterns/S) → 382 days for 1 million chips



Test Pattern #	Test Pattern (I5, ..., I2, I1)	Output (OG)
1	00000	0
2	00001	0
...		0
2^5	11111	1

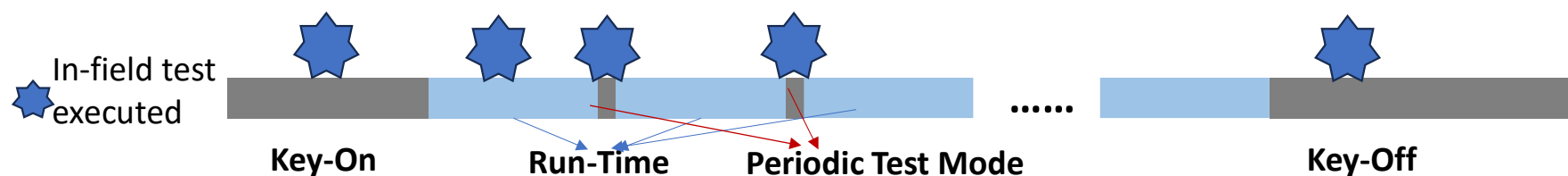
6×2^5 Test patterns → 0.00019 Seconds per chip (1 MHz tester: running 10^6 patterns/S) → 192 Seconds for 1 million chips

A Deep Dive of Manufacturing Test Flow



In-Field Test Introduction

- Service quality of silicon-based systems is critical to business, e.g.,
 - Data center/cloud provider (as-a-service business model) → zero tolerance for unplanned downtime
 - Automotive/industrial IoT → zero tolerance for safety hazards
- **In-field Test:** to detect if any such service failures that are beyond the acceptance level
- Characteristics of In-field Test:
 - Resource-constraint; no ATE available (unlike manufacturing test); more DFT (e.g., BIST) needed - to cover later in this lecture
 - Test to be performed in “Key-on”, “Key-off”, “Periodic Test” or “Run-Time” mode

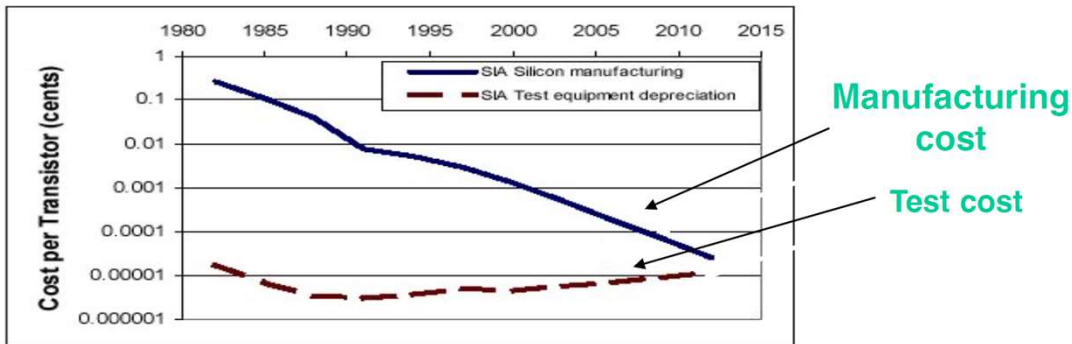


- Require minimal impact on system normal operations (performance, power etc.)
- Relatively new field, especially in autonomous driving, data center domains

DFT: WHY

- Manufacturing test cost is high (and spiking)
 → Need to reduce cost: test pattern ↓, Test time ↓....

IEEE International Technology Roadmap for Semiconductors



1 second test time
 reduction per chip → 11.6
 days test time reduction
 for 1 million chips



Test Costs Spiking

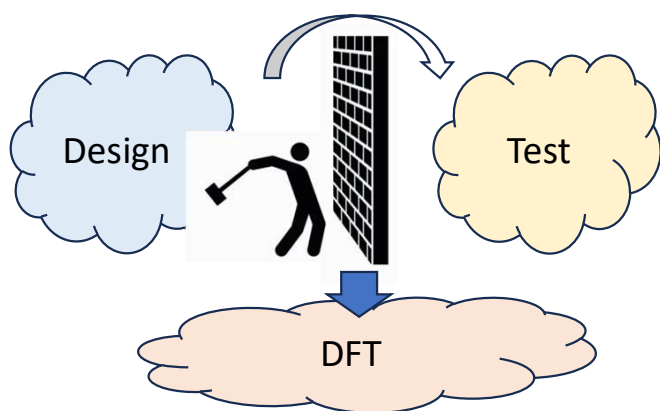
Use of more complex chips in safety- and mission-critical markets is changing pricing formulas for manufacturing.

MARCH 10TH, 2020 - BY: ED SPERLING

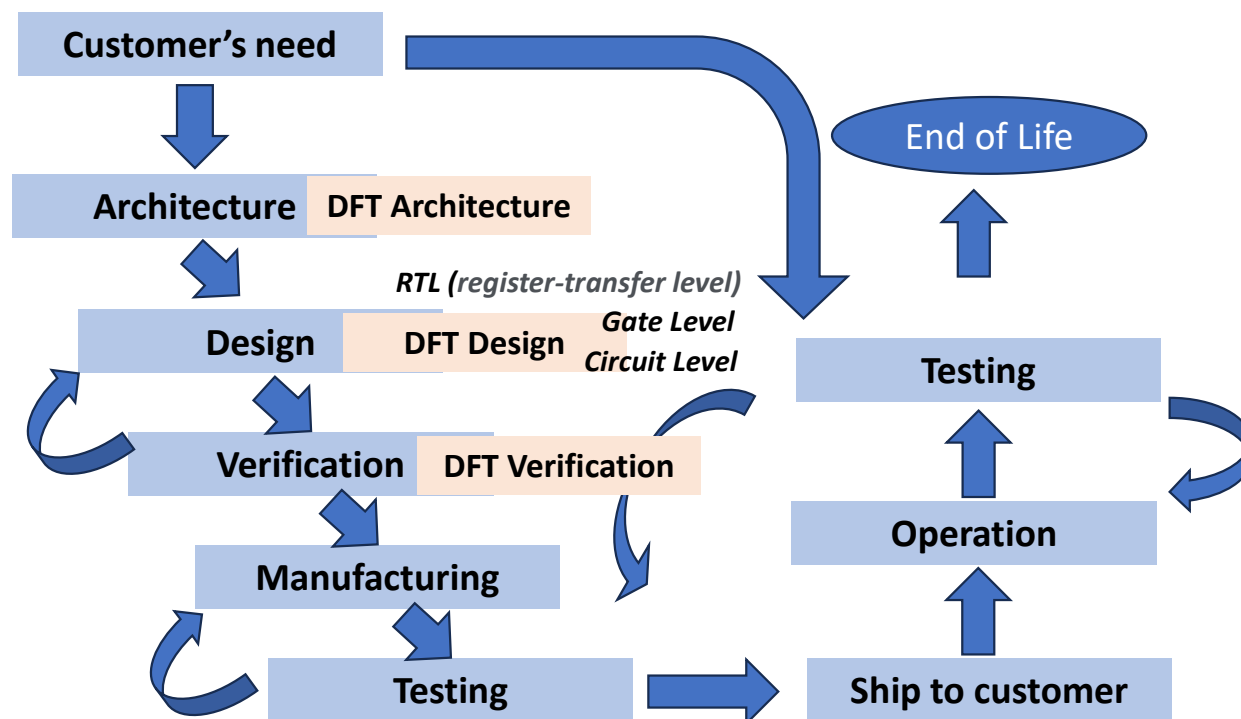

Strong Need of DFT
 (Design for Testability)

- In-field test challenges and complexity
 → Need to support efficient in-field test solutions

DFT in Silicon Lifecycle



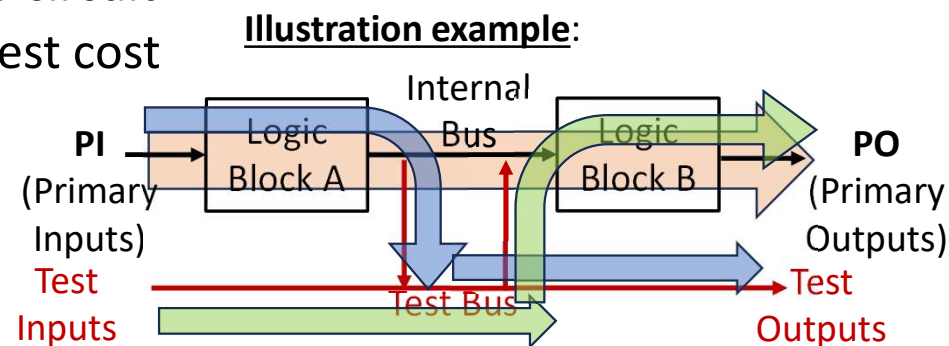
VLSI Chip Lifecycle



DFT: WHAT

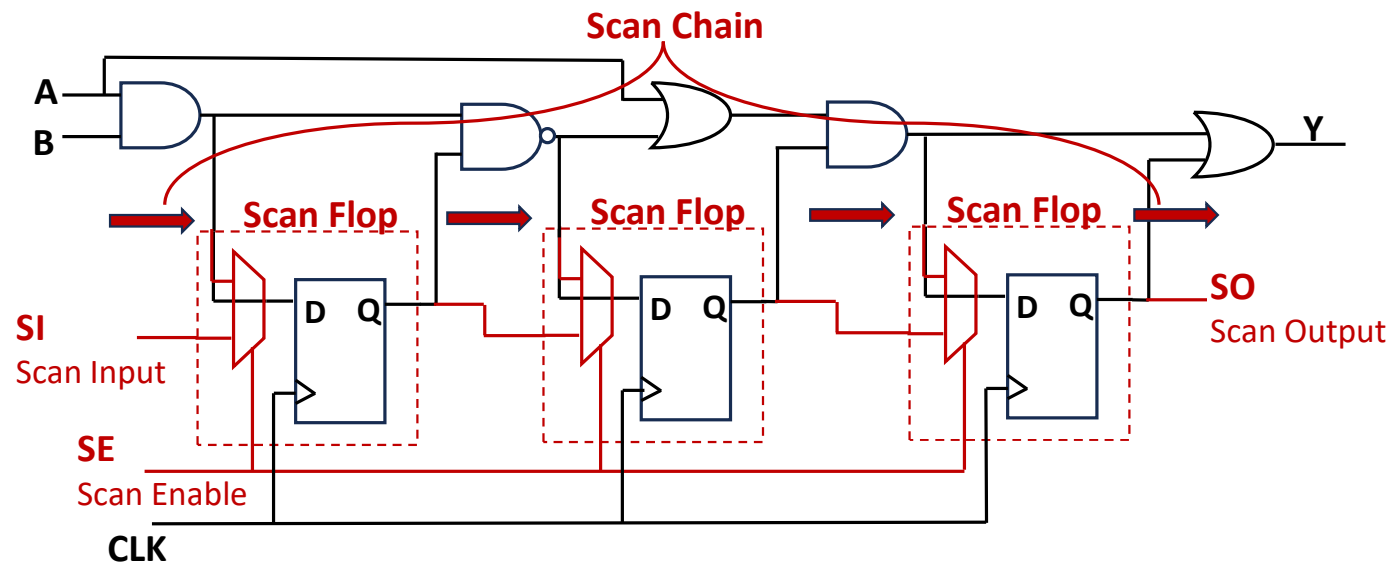
- Design for Testability (DFT): hardware design style or added hardware to reduce test solution's cost and complexity
- Testability = Controllability + Observability
 - Controllability: the ability of applying the test stimulus to DUT (Device Under Test)
 - Observability: the ability of observing test response (including internal states) of DUT
- DFT purposes:
 - Add controllability and observability to circuit
 - Ease the test process and reduce the test cost

Test HW applies test to blocks A and B individually via the test bus; avoid test generation for combined A and B blocks → reduce test generation cost



DFT Example: Scan

- Scan is the internal modification of the design's circuitry to increase its testability.
 - Scan chains are the elements used to shift-in and shift-out test data
 - Scan-based testing is a primary solution for logic testing

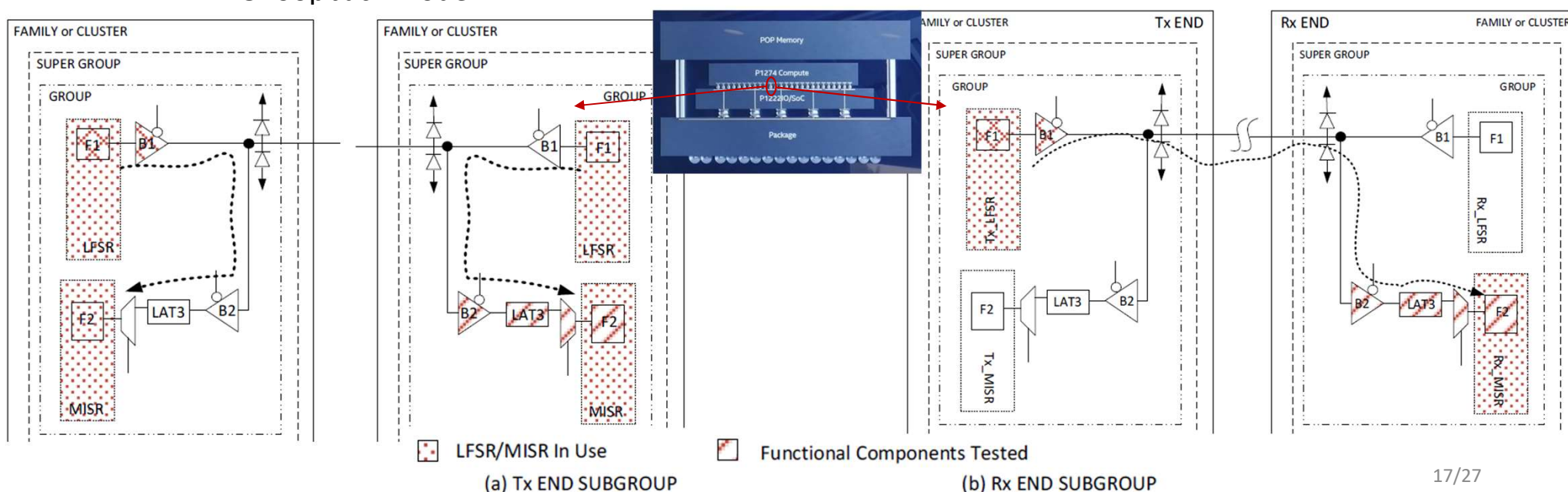


DFT Example: BIST

- BIST (Built-In Self Test) – reduce ATE dependency
 - Example: Chiplet Die-to-Die IO interconnect BIST
 - LFSR (linear-feedback shift register): Pseudo random test pattern generator
 - MISR (multiple-input signature register): test pattern comparator
 - IO loopback mode

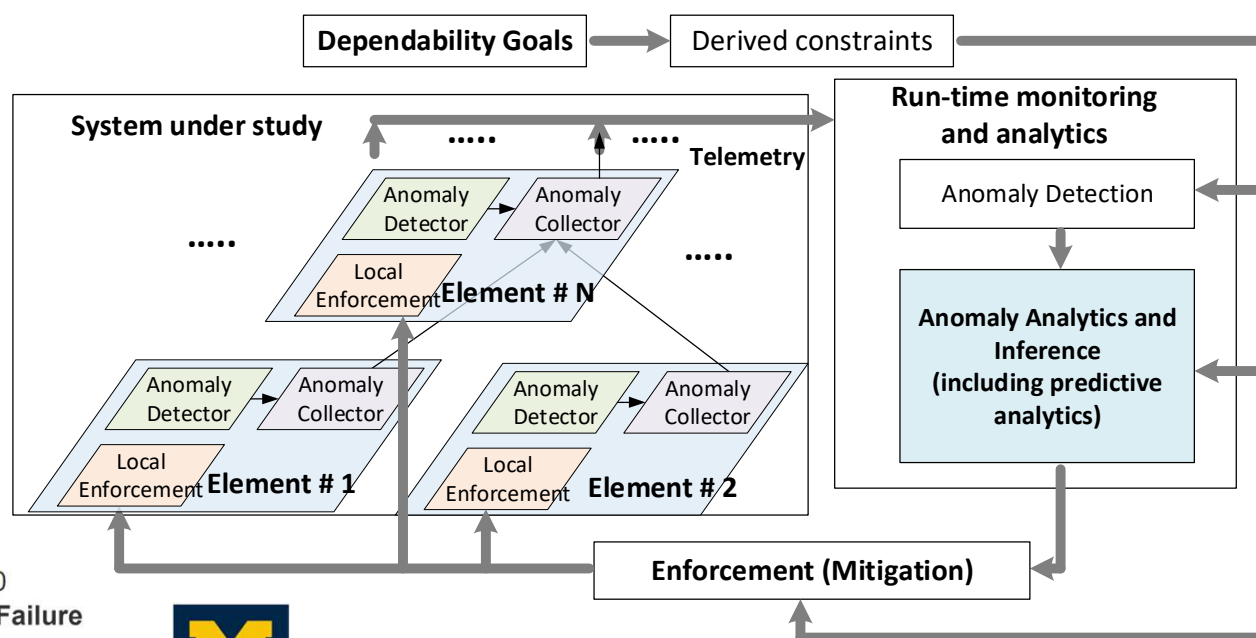
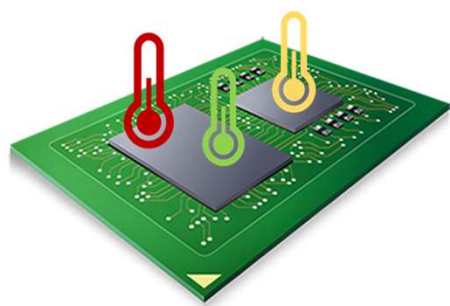
Internal I/O Testing: Definition and a Solution

Sreejit Chakravarty, Fei Su, Indira A Gohad, Sudheer V Bandana, B S Adithya, Wei-Ming Lim
 Email: {sreejit.chakravarty, fei.su, indira.a.gohad, sudheer.v.bandana, b.s.adithya, wei.ming.lim}@intel.com
 Intel Corporation



DFT Example: Intelligent Sensor Based DFT

- On-chip sensors & analytic engines
- New solutions for manufacturing testing & in-field testing



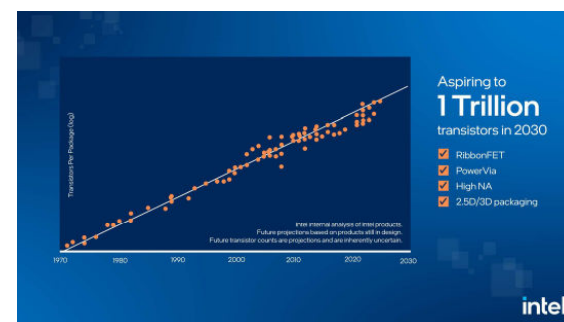
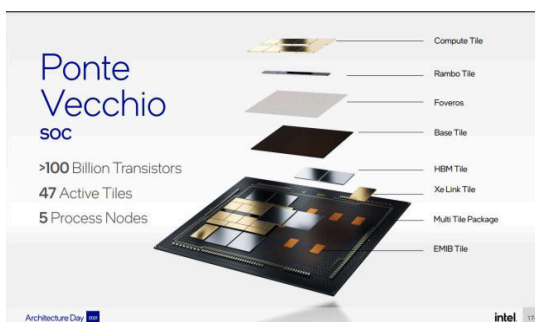
Part II: Advances in AI-for-DFT

Agenda

- ❑ The Growing Complexity Challenge
- ❑ What is AI-for-DFT?
- ❑ Deep Dive
 - ❑ Deep Dive 1: AI for Fault Analysis & Diagnosis
 - ❑ Deep Dive 2: AI for Intelligent Test Pattern Generation
- ❑ Future Outlook

The Growing Complexity Challenge

- ❑ Moore's Law & Advanced Packaging: Complexity is shifting and exploding.
- ❑ Billions of transistors, 3D integration, heterogeneous systems → traditional methods inadequate.
- ❑ Paradigm shift needed: from human-driven to AI-driven methodologies.

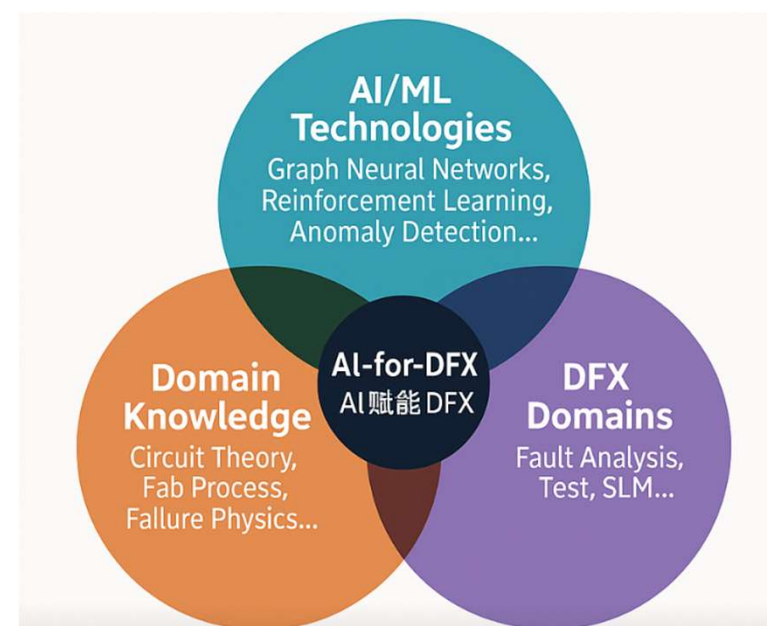


Source: [The Future of Moore's Law](#), Intel Investor Meeting 2022

Source: [Intel Architecture Day 2021](#)

What is AI-for-DFT?

- ❑ DFT & DFX (Design for Excellence): includes DFM, DFR...
- ❑ DFX pain points:
 - ❑ Fault Analysis: root cause elusive
 - ❑ Test Generation: coverage gaps costly
 - ❑ In-field Reliability: unknown failures.
- ❑ AI-for-DFT: using AI/ML to predict, optimize, and automate across silicon lifecycle.



Deep Dive 1: AI for Fault Analysis

- ❑ Problem: Post-silicon debug slow, expert-dependent.
- ❑ AI for Fault Analysis Examples
 - ❑ GNN Solution: Treats netlist as graph → maps failure signatures to fault locations.
 - Benefit: Reduce root-cause isolation time from weeks to hours.
 - ❑ Anomaly Detection: unsupervised learning on volume test data to find subtle, novel failures.
 - ❑ Fault criticality analysis for AI Chip (**Deep Dive**)

Functional Criticality Classification of Structural Faults in AI Accelerators

- Analyze functional criticality of structural faults in systolic array-based AI accelerators
- Fault criticality analysis can guide test grading and quality assessment
- Develop machine-learning (ML) based technique for fault-criticality classification
 - Avoid **computationally expensive** brute-force fault simulations

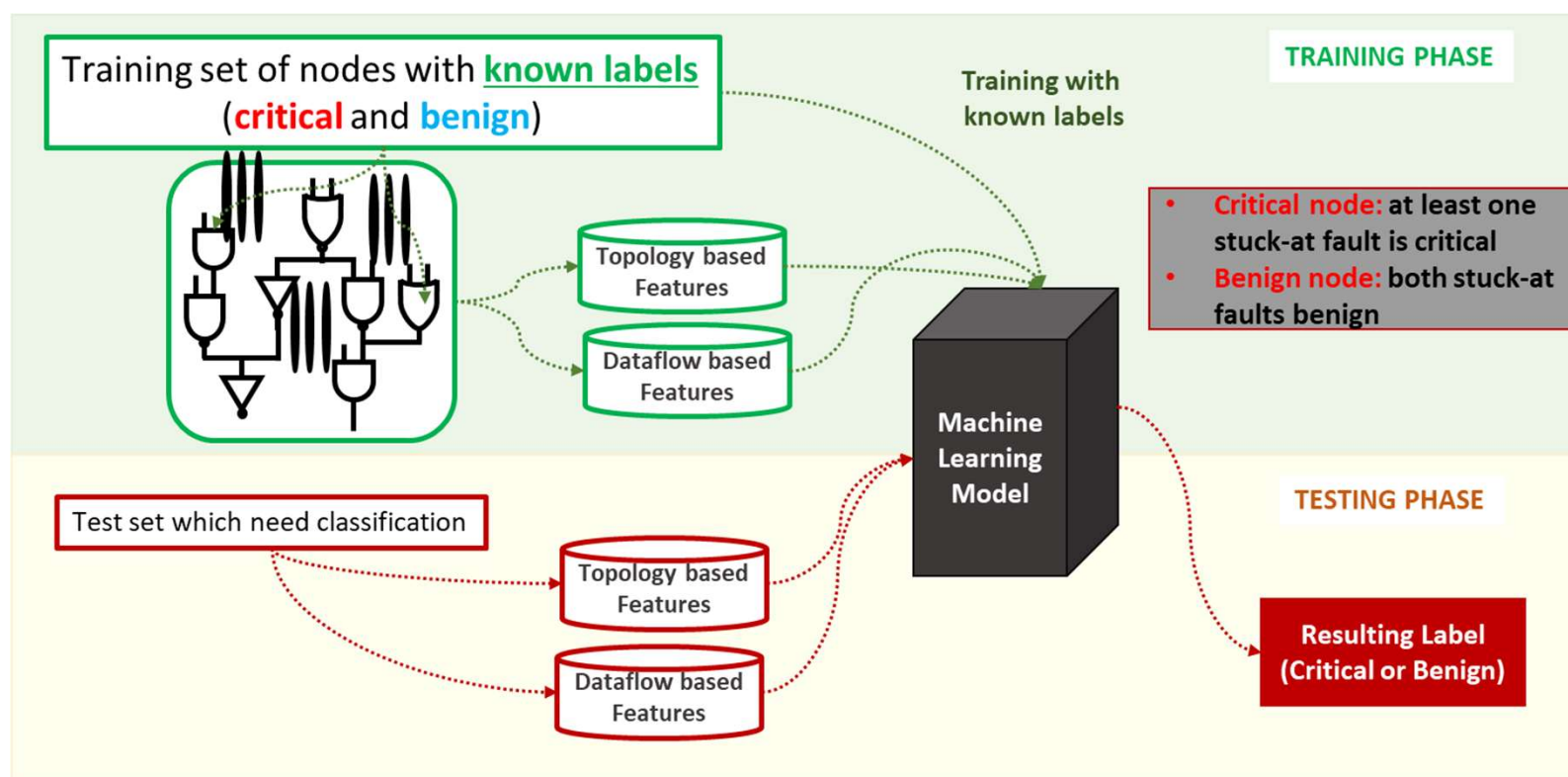


<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9325272>

A. Chaudhuri, J. Talukdar, F. Su, and K. Chakrabarty IEEE ITC 2020; IEEE TCAD 2022

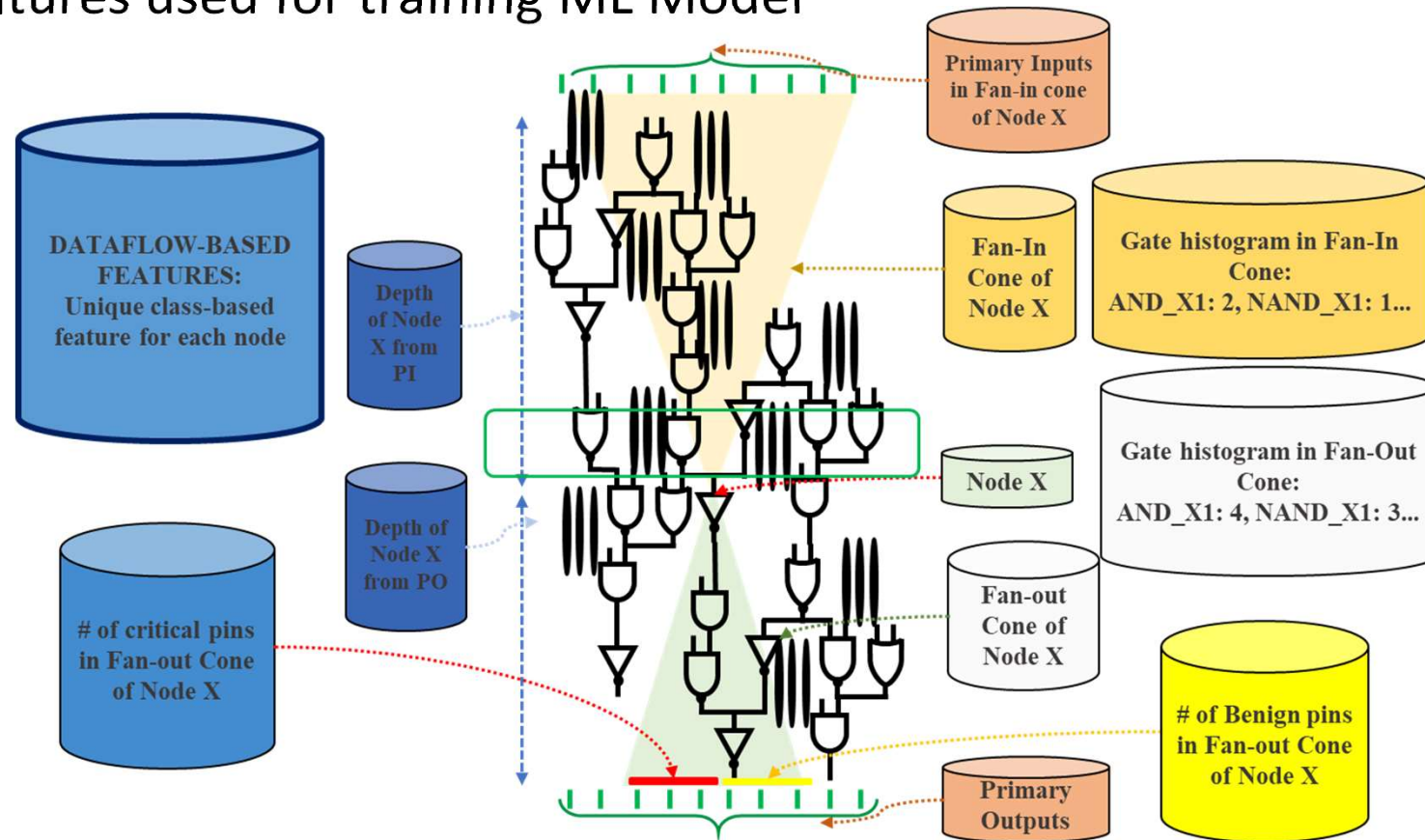
Functional Criticality Classification of Structural Faults in AI Accelerators

- Fault-criticality Assessment using ML



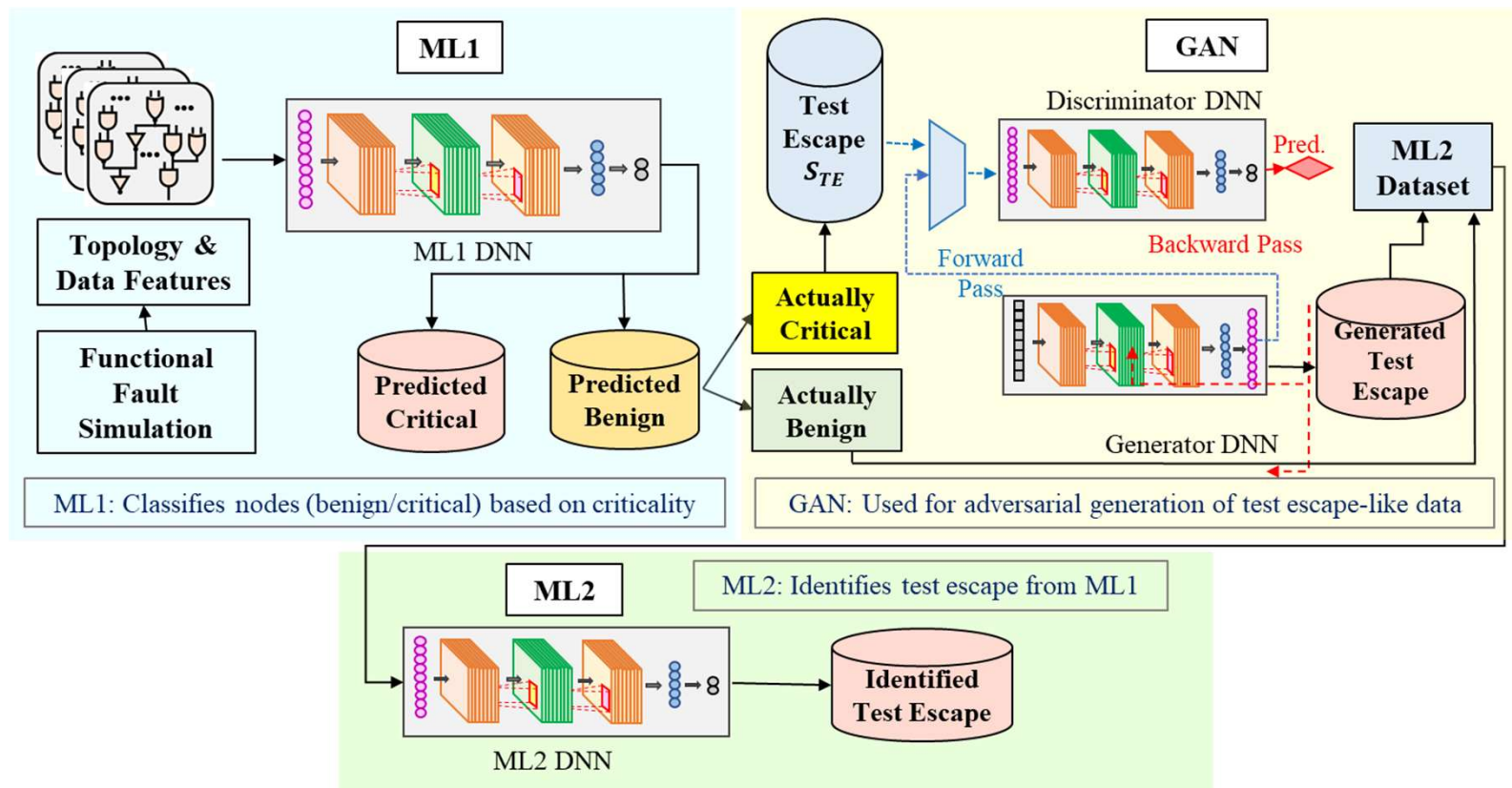
Functional Criticality Classification of Structural Faults in AI Accelerators

- Features used for training ML Model



Functional Criticality Classification of Structural Faults in AI Accelerators

- Aggregated GAN-based Two-Tier ML Training (ML1 → GAN → ML2)



Functional Criticality Classification of Structural Faults in AI Accelerators

- Two-tier ML method is effective in evaluating criticality of internal faults in PE
- Topology and data-based features: good representation of internal connectivity and dataflow of the PE netlist
- Training a GAN to generate more test escape-like data significantly reduces misclassification in ML2
- Aggregated-GAN based fault-criticality assessment flow identifies catastrophically critical faults with negligible test escape

Deep Dive 2: AI for Intelligent Test Pattern Generation

- ❑ Problem: Test pattern volume and application time unsustainable
- ❑ AI for Intelligent Test Pattern Generation Examples:
 - ❑ RL for ATPG: learns to generate effective patterns maximizing coverage with fewer vectors
 - Benefit: reduce TDV and test time without quality loss.
 - ❑ Predictive Models: early prediction of test coverage and untestable areas → guide DFT insertion
 - ❑ GAN for FuSa test pattern generation (**Deep Dive**)

FuSa Test Pattern Generation for AI Chips

- AI Chip memory circuits highly vulnerable to faults due to various reasons like aging, latent defects, single event upsets, etc.
- Such faults are highly detrimental to the classification accuracy of the DNN accelerator, leading to the crucial Functional Safety (FuSa) violation
- To detect such violations in mission mode, we propose to generate a set of functional test patterns by leveraging the concept of Generative Adversarial Networks (GANs)

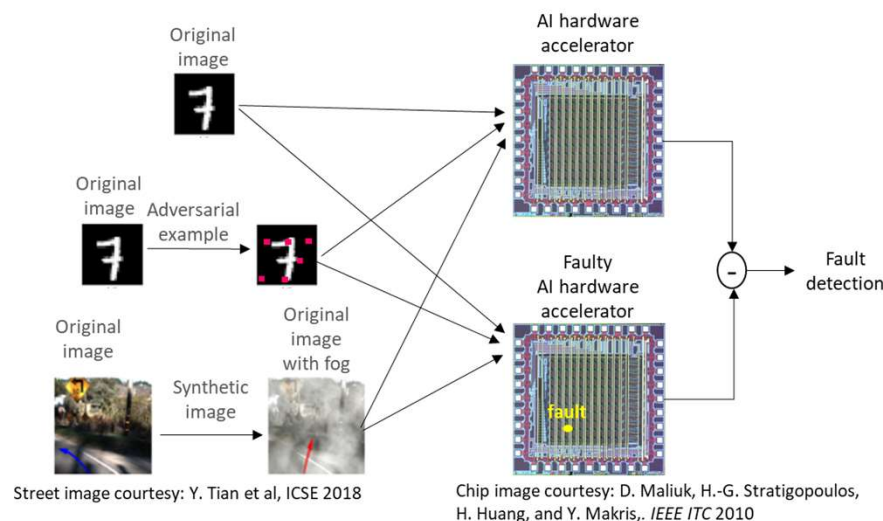
Joint work with S. Kundu/K. Basu (UT-Dallas), S. Banerjee/A. Raha/S. Natarajan (Intel)



FuSa Test Pattern Generation for AI Chips

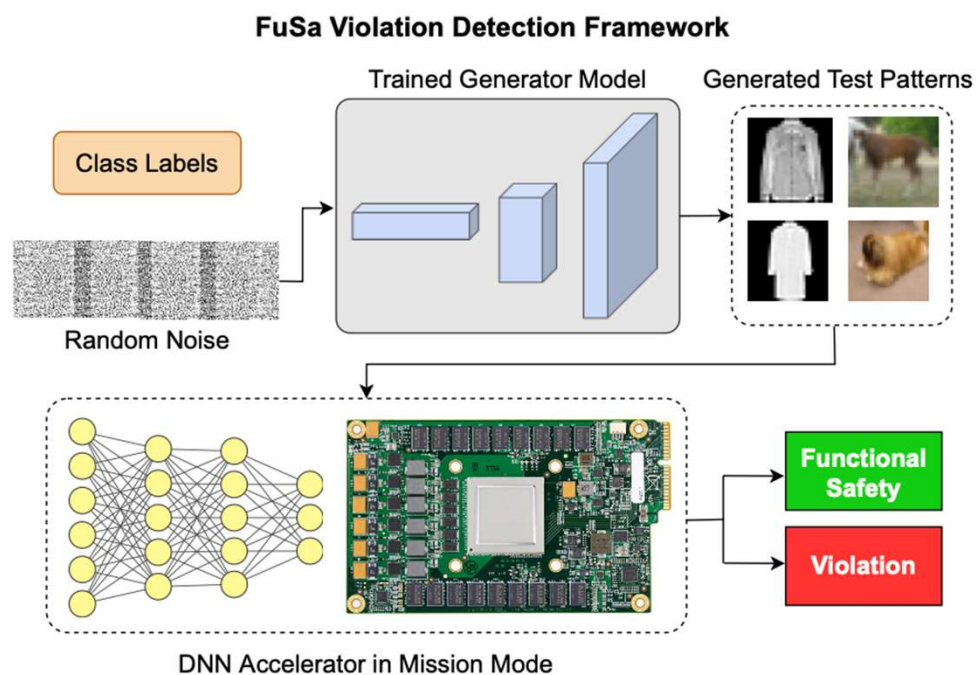
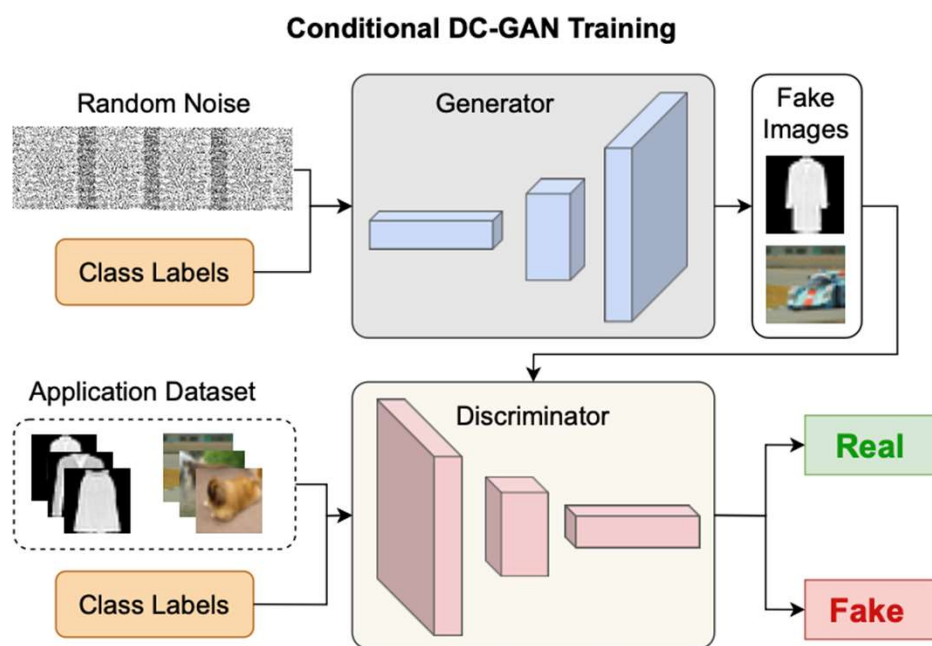
- Challenge to utilize ATPG patterns to detect FuSa violations in a DNN accelerator.
 - Patterns generated from ATPG cannot be used in mission mode.
 - Structural patterns do not represent a specific image & are inefficient in detecting FuSa violation
- Functional test generation for FuSa → generate inputs, e.g., images, that are capable of sensitizing the fault and propagating its effect to the output, leading to a different AI service performance with respect to fault-free ones (potential FuSa violation)

- Example of test patterns:
 - Original data sample selected from training or testing dataset
 - Adversarial data generated from original data
 - Synthetic data

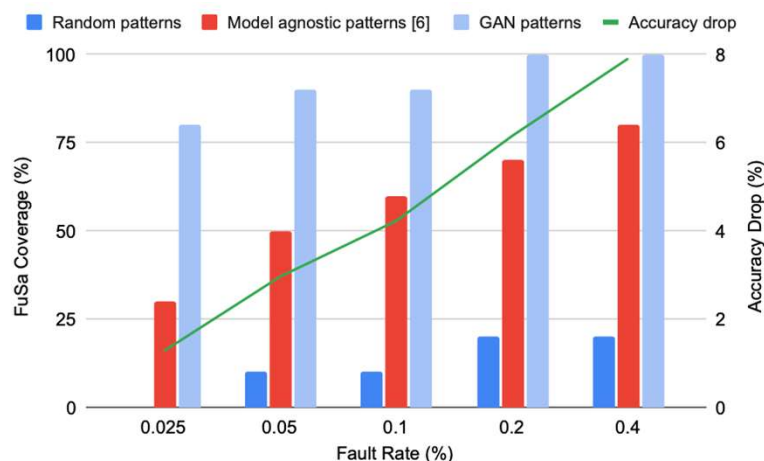


FuSa Test Pattern Generation for AI Chips

- ❑ We need to obtain a test pattern set with low cardinality and high coverage
- ❑ Proposed GAN-based test pattern generation



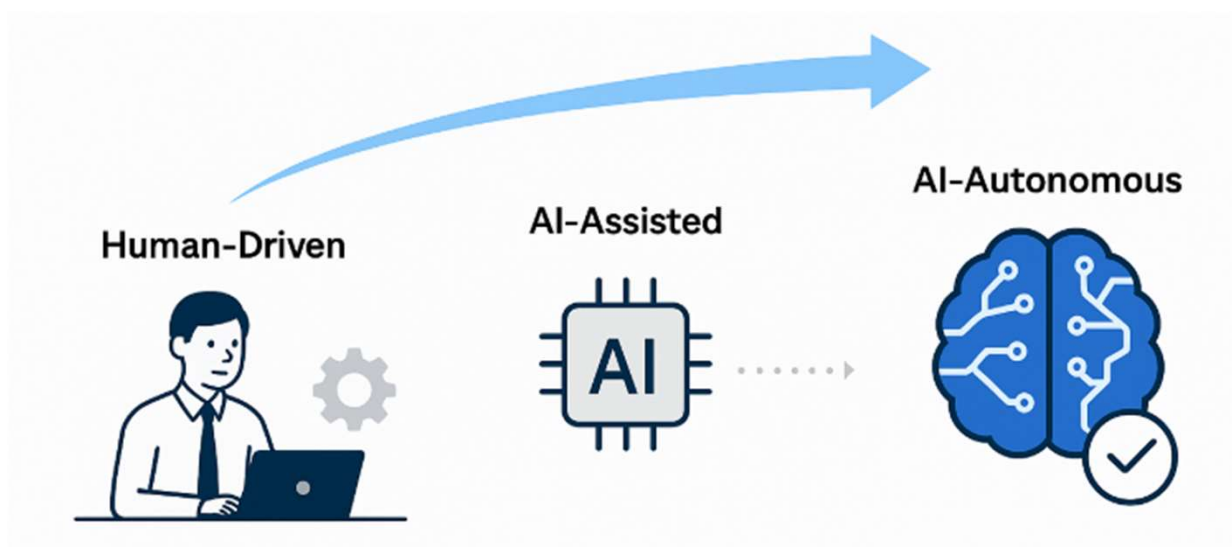
FuSa Test Pattern Generation for AI Chips



- Leverage the concept of GAN to generate FuSa functional test pattern
- GAN-generated test patterns significantly improves the FuSa violation detection rate in DNN accelerators
- Framework can be extended to datasets other than images

Future Outlook

- ❑ AI-for-DFX is essential to manage complexity and ensure quality in chiplet era.
- ❑ The Future: Self-Healing Chips, Autonomous DFX Agents.
- ❑ **Call to Action:** Now is the time to build your data strategy and invest in AI-for-DFT



Part III: Advances in Silicon Lifecycle Management

OUTLINE

WHY need SLM (Silicon Lifecycle Management) ? a.k.a. what are pain points in the age of Heterogenous Integration?



What are SLM solutions: Heterogenous Integration age challenges and opportunities?



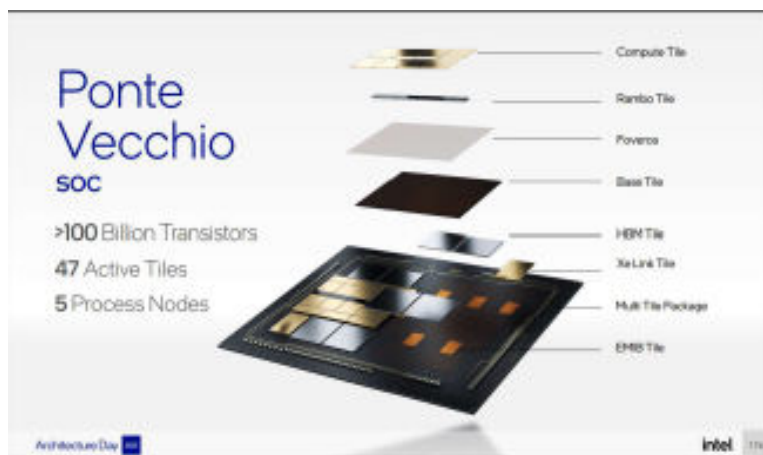
How do SLM solutions address pain points?
Example use cases

HETEROGENEOUS INTEGRATION AGE

Advanced chip integration technology, e.g. Chiplet, is critical for overcoming the limitations of Moore's Law

It fundamentally relies on **heterogeneous integration** (异构集成), which allows different functional components to be packaged together as a single system

Its importance is magnified by China's lack of access to EUV lithography, making it a **strategic national priority**



Source: [Intel Architecture Day 2021](#)

A PARADIGM SHIFT IN QUALITY CONTROL

Key Challenge Areas:

1. Interconnect Integrity & Signal Integrity:

- Micro-bump/bonding defects (voids, misalignment) and interfacial contamination.
- Increased parasitic effects and crosstalk, leading to signal degradation and timing errors.

2. Thermal & Mechanical Reliability

- Warpage and stress due to CTE mismatch between dissimilar materials
- Thermal hotspot management and delamination risks under power cycling

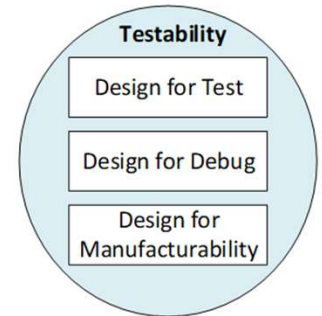
3. Test & Diagnostics Limitations

- Lack of internal node observability, making post-assembly fault isolation extremely difficult
- Inadequate pre-bond test coverage for complex dies, resulting in latent defects

4. Known Good Die (KGD) & System Yield

- The final system yield is a product of individual die yields, making KGD availability and quality paramount
- Lack of standardized KGD specifications and testing protocols across the supply chain

- Quality must be **designed-in** and **co-optimized** across design, process, and test.
- Requires new standards for KGD, new test tools, and system-level quality metrics.

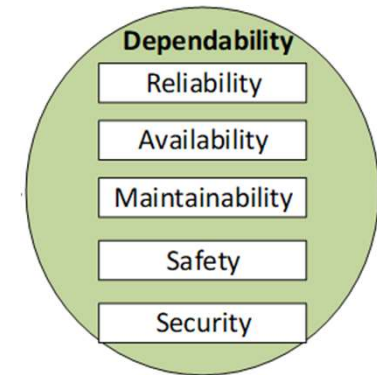


RESILIENCE: THE NEW FRONTIER

Resilience is key to

- data center/cloud provider (as-a-service business model) → zero tolerance for unplanned downtime
- automotive/industrial IoT → zero tolerance for safety hazards

Dependability: ability to avoid such service failures that are beyond the acceptance level



Example: Data center SDE (Silent Data Error)/SDC (Silent Data Corruption) issues caused by early-life failure, degradation or end-of-life wear-out

Google and Meta papers in 2021

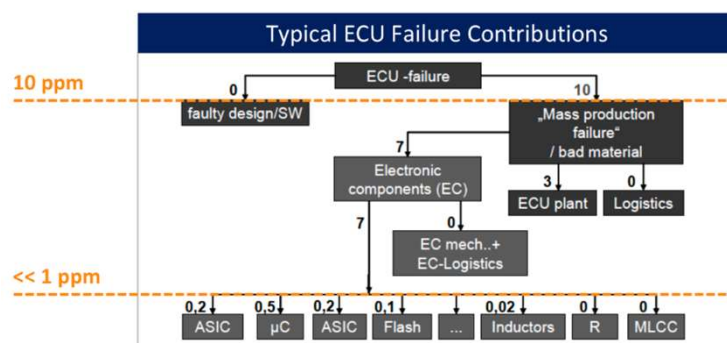
- Facebook/Meta “Silent Data Corruptions at Scale”
- Google “Cores that Don’t Count”

...WITH HIGHER QUALITY REQUIREMENT

0 DPM requirement: DPM (Defects Per Million) $\rightarrow$ DPB (Defects Per Billion)

Example: Automotive – electronics failure

- Audi reports “about 7,000 semiconductor devices in its high-end models, and it turns out about 4,000 cars every day” (Source: Semiconductor Engineering 2019).
- A failure in the range of 1 DPM $\rightarrow$ 28 defective cars per day



**Lower DPM requirements $\rightarrow$
More difficulty to meet them**

Source: Semiconductor Engineering/ Optimal+

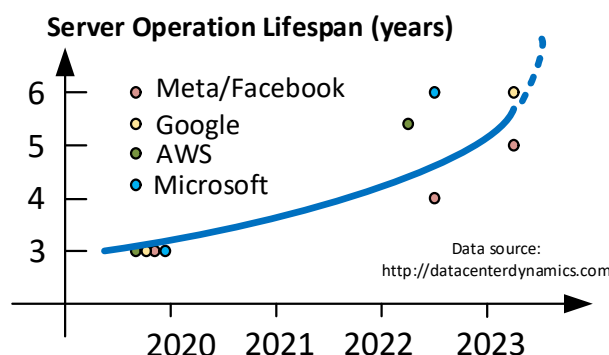
Similar trend occurs in cloud/hyperscaler marketing segment

...WITH EXTENDED LIFETIME

Automotive/industrial: 10-15 years

Cloud/Hyperscalers: New Challenge

**Longer lifetime → More chance
to encounter the dependability
issues!**

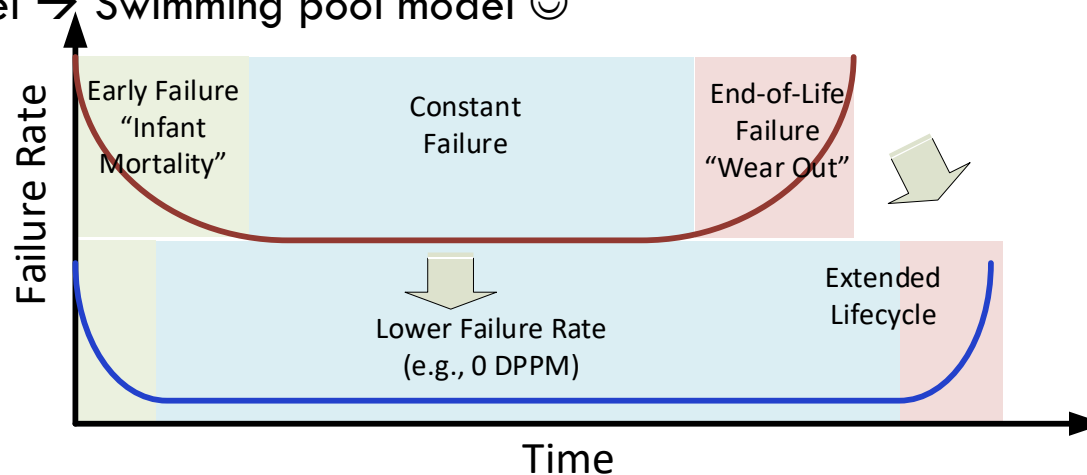


"We expect [the change] to result in a reduction of depreciation of approximately **\$3.4** billion for the full fiscal year 2023 for assets in service as of December 31, 2022, recorded primarily in cost of revenues and research and development (R&D) expenses," Google's parent Alphabet said in an earnings release.

In its earnings call this month, Meta announced that it would extend the expected life of servers and some network equipment to five years. This, it said, would save around **\$1.5bn.**

WHY → WHAT IS NEEDED

Bathtub model → Swimming pool model 😊



These **pain points** together drive a need of comprehensive solution of managing (not just testing) silicon/system **testability and dependability issues** throughout its life cycle
 → Silicon Lifecycle Management (**SLM**) solution

HETEROGENEOUS INTEGRATION AGE CHALLENGES

Existing solutions: Add more manufacturing test efforts; Add more guard band to designs; Add more redundancy; Invest more in RMA...

*What are the challenges rendering existing solutions **not scalable enough to fully address** the increasing needs of SLM across industry (silicon vendors and customers)?*



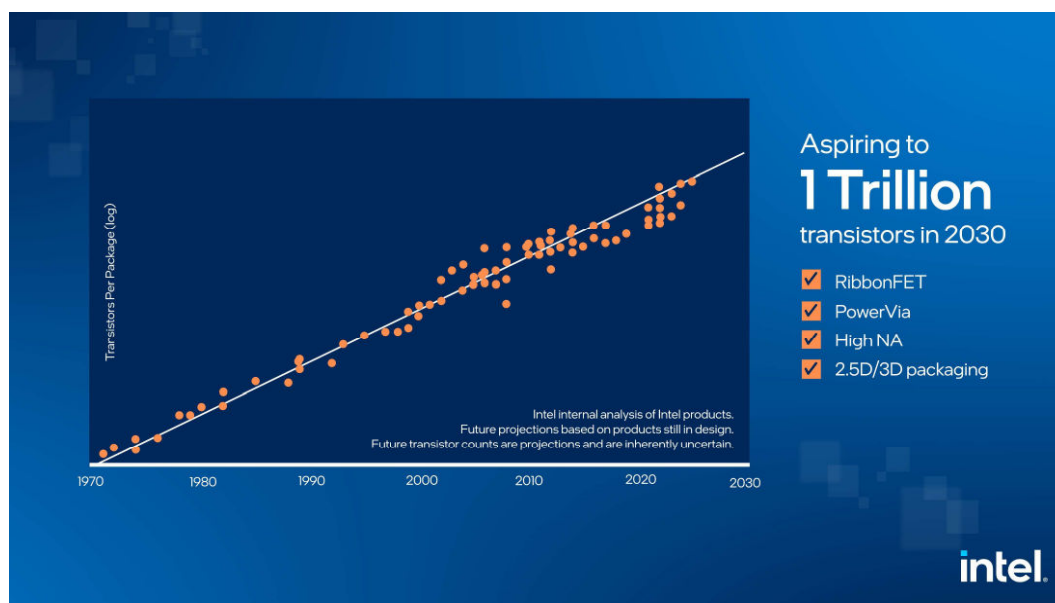
Technology level challenges

Chip level challenges

System level Challenges

CHALLENGES — TECHNOLOGY LEVEL

Technology scaling — shrinking of process node → more failure modes

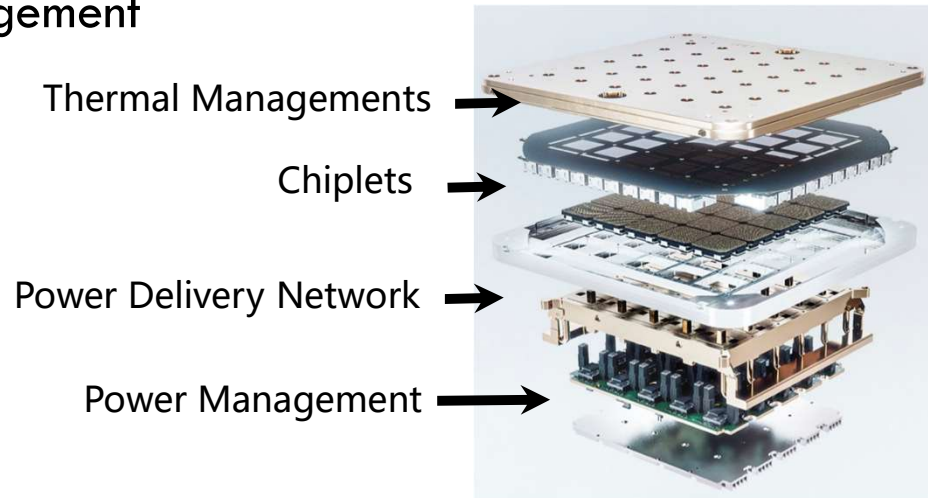


Source: [The Future of Moore's Law](#), Intel Investor Meeting 2022

CHALLENGES — CHIP LEVEL

Integrating multiple chips into a larger system demands mastery over a complex set of disciplines—from process and reliability to power, thermal, and interconnect design

This very convergence of challenges is pushing the boundaries of what's possible in chip health management



Courtesy: Tesla Dojo
Compute Chip Module

CHALLENGES — SYSTEM LEVEL

Growing system complexity of HW-SW co-optimization

- Need system-level optimization
- Need to monitor performance of applications, e.g., root cause analysis of sub-optimal behavior in on-chip communication or inter-processor synchronization

Complexity of emerging workload

- ML workload
- Adaptive workload – specially real-time management needed

At-scale application

Saturating effectiveness of manufacturing screen

NEW SLM ARCHITECTURE

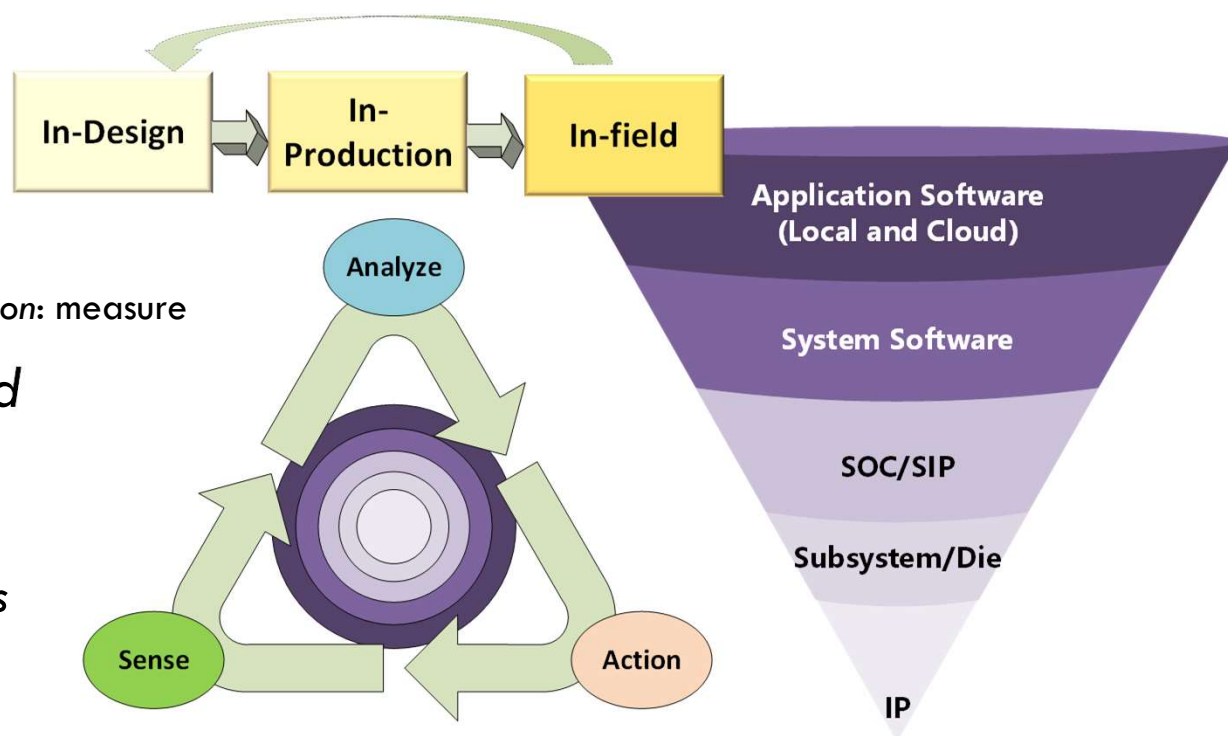
Silicon Lifecycle Management (SLM) refers to the holistic management and continuous tracking of a chip's performance, health, reliability, and security throughout all stages

To address multifold challenges of SLM →

Telemetry – From *Tele*: remote + *metron*: measure

A holistic telemetry-based architecture for SLM:

- Hierarchical arch
- Span across lifecycle phases



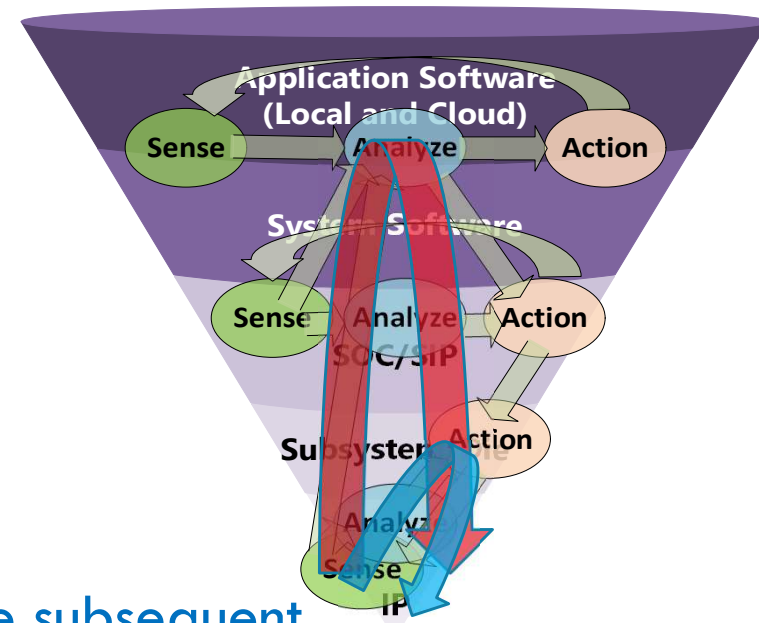
NEW SLM ARCHITECTURE

Values of new SLM architecture:

- Intelligent Sensing
 - Continuous monitoring (dynamic env, workload activated issue etc.)
 - Sensing alone cannot address all use models
- *“Intelligence is the ability to adapt to change” – Stephen Hawking*
- Data security and privacy concerns must be addressed
- Data volume challenges
- Actionable Intelligence enabled by run-time analytics
 - Ability to quickly identify and communicate failures/threats
 - Real time, closed loop feedback for performance tuning

The ability to infer actionable intelligence and drive subsequent actions and continuous learning will be a differentiator

Example:



DATA TO SENSE

Physical data

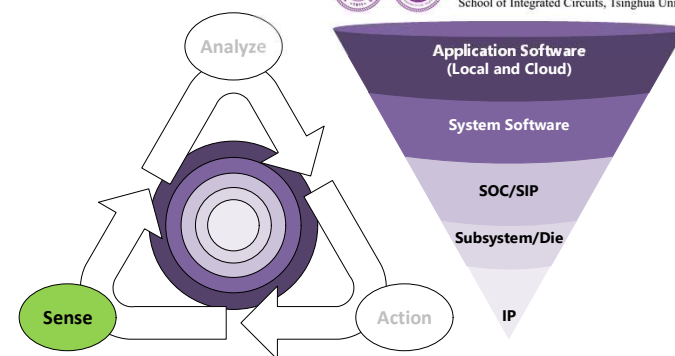
- Silicon PVT (Process, Voltage, Temp): IP, die, SOC
- Platform/environmental (Voltage, Temp, Mechanical etc.)
- Structural data: in-field scan

Functional data

- HW Performance monitor
 - Critical-path margin
 - Power-related performance
- SW performance monitor

Integrity data

- Machine check data
- In-field test data
- Security data



Example of data measures

Platform Stack	Example Product Health Measures
Application Software	Foreground applications: name, ID, open & close times, CPU utilization
Services and Features	Feature present/enabled, configuration, application usage
Operating Systems	Event logs, e.g., bug check , on and off times, system utilization
Firmware and Drivers	Corrected error events , system configuration, CPU processor name
CPU	Temperature , power states, voltage
Chipset & Peripherals	Temperature

ANALYZE THE DATA

What to analyze?

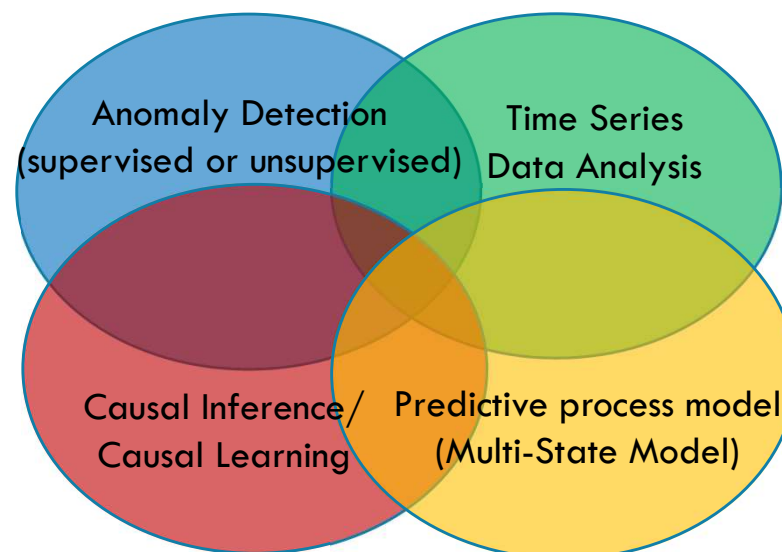
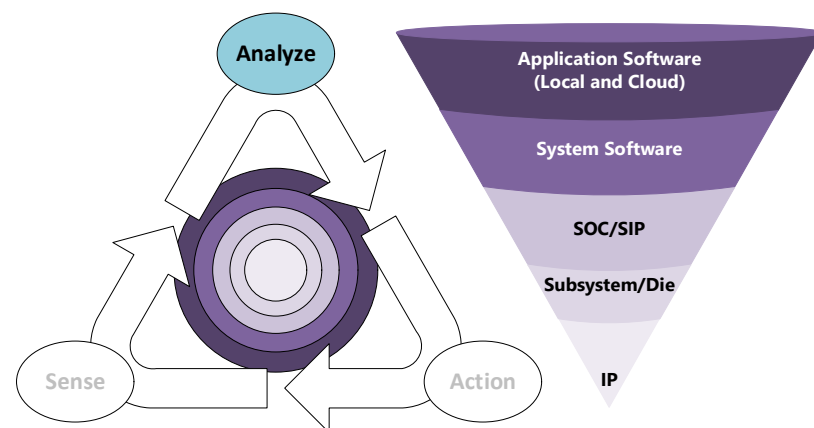
- Detect: “Is there an issue?”
- Scoping: “Is it a local issue or a global issue?”
- Triage: “Who should work on the issue?”
- Debug: “What is the root cause of the issue?”
- Validation: “Did the issue fix work?”
- Prediction: “Can we predict/prevent the issue?”

Where to analyze?

- In-sensor or Near-to-sensor
- SOC or die
- Platform level
- Cloud (remote)

How to analyze?

- Domain expertise
- ML/AI approach to take. Example →



ACTION BASED ON DATA

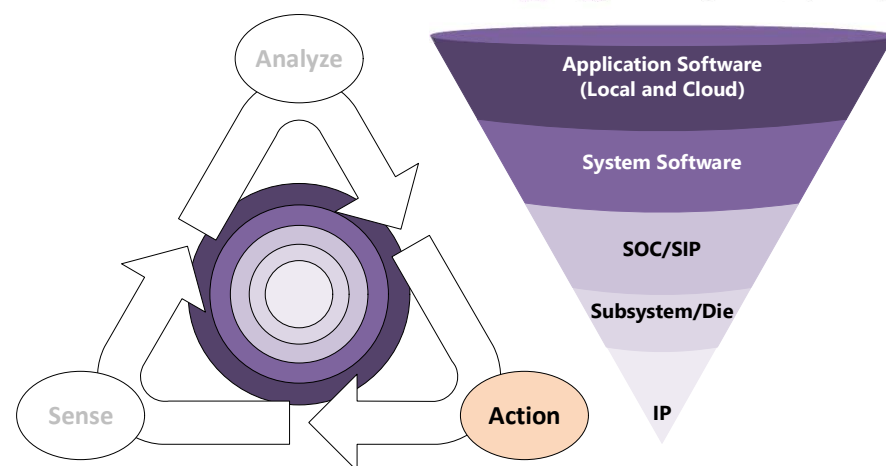
Different types:

- Reactive action
- Proactive action

Domain:

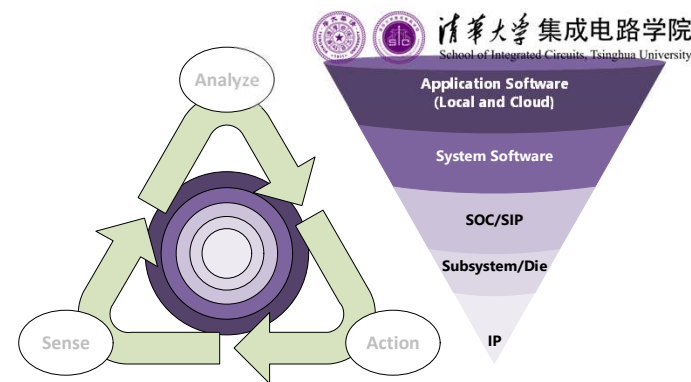
- PnP (Power and Performance) run-time optimization
- Safety/Security mitigation
- Predictive Maintenance

Repair (mask & spare)

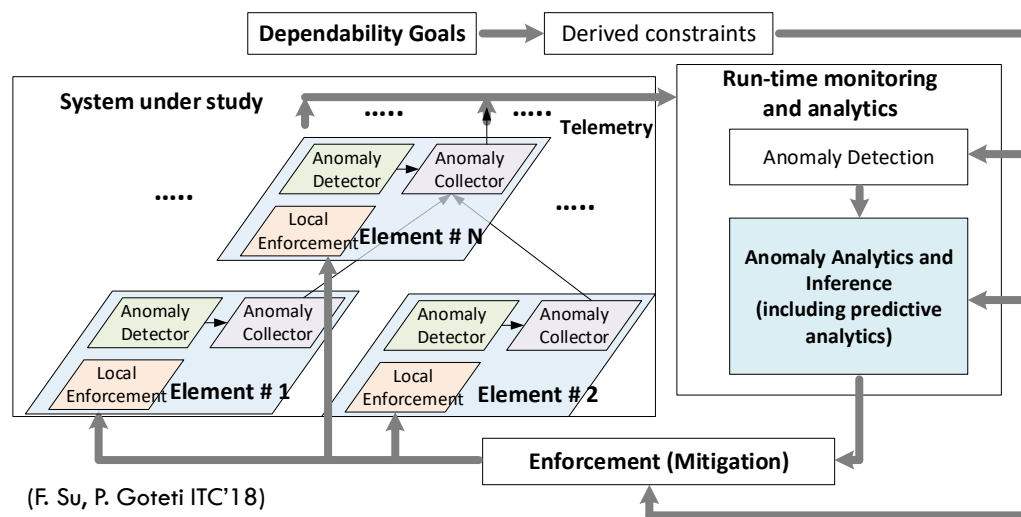


TELEMETRY

Need a holistic telemetry architecture and platform stack to connect “Sense”, “Analyze” and “Action” blocks and support communications among them

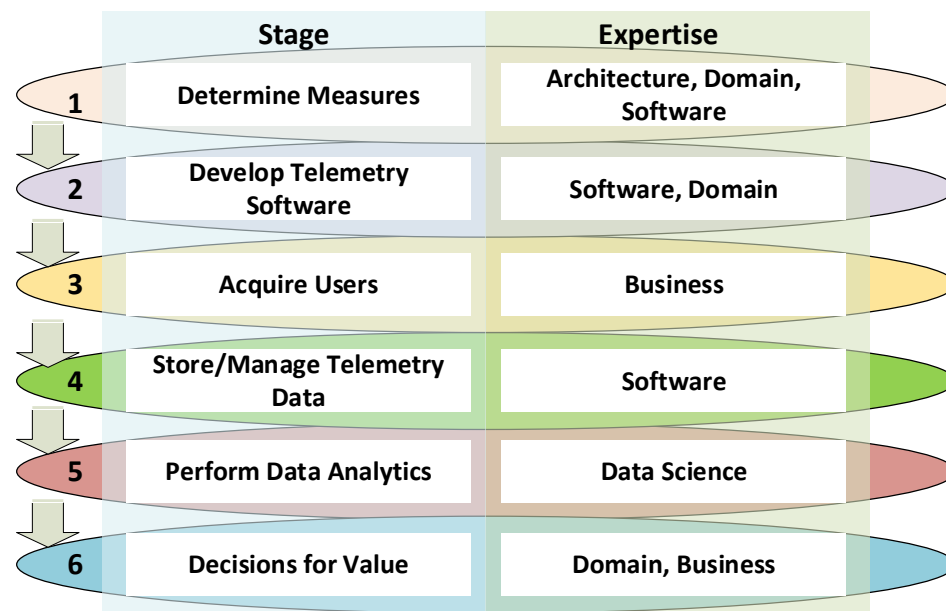


Anomaly Detection based Telemetry Architecture



(F. Su, P. Goteti ITC'18)
 (F. Su, P. Goteti, M. Zhang, ITC'19)
 (R. Kundu, F. Su, P. Goteti JETTA'19)
 (F. Su, P. Goteti, M. Zhang, ITC'20)

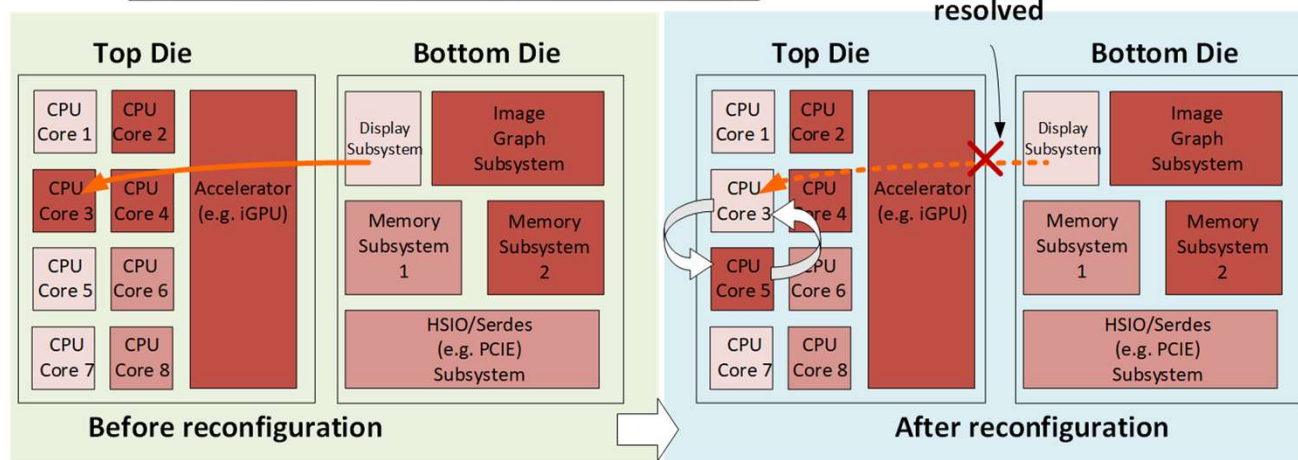
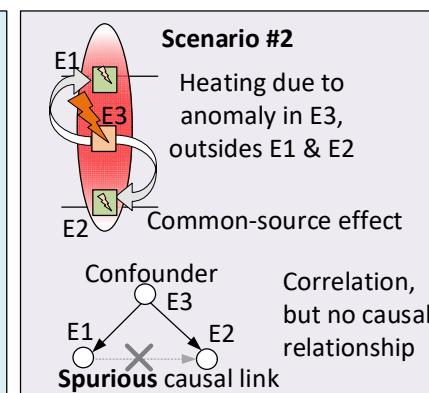
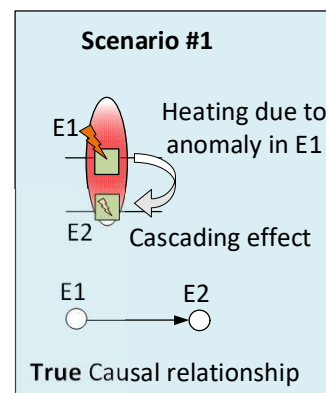
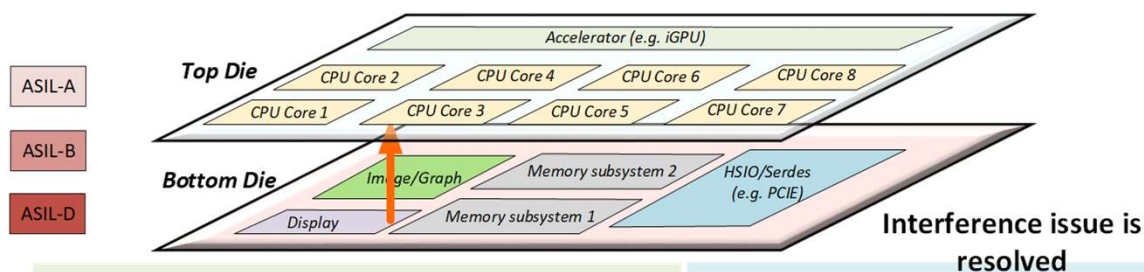
Telemetry Platform Stack



(F. Su, et. al IEEE D&T 24)

USE CASE EXAMPLE

Automotive: Freedom from Interference (FFI) Verification



Reconfiguration: re-allocating ASIL D function from CPU Core 3 to Core 5

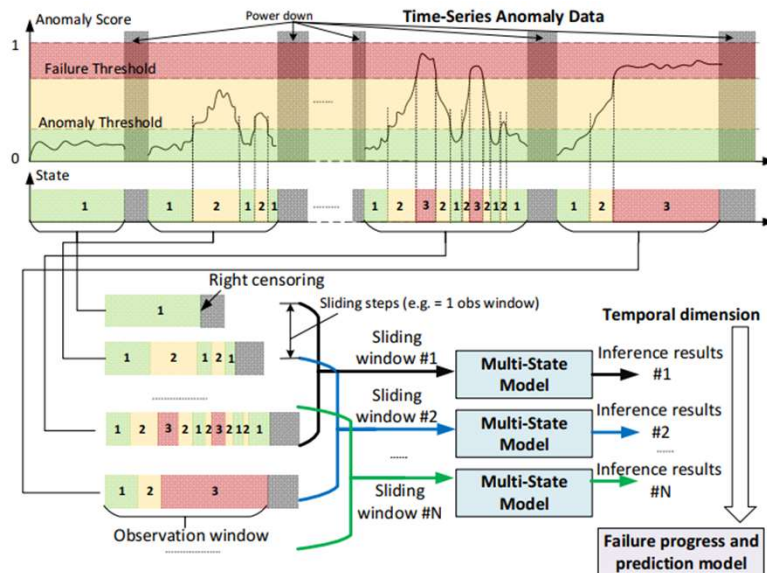


(ITC'19)

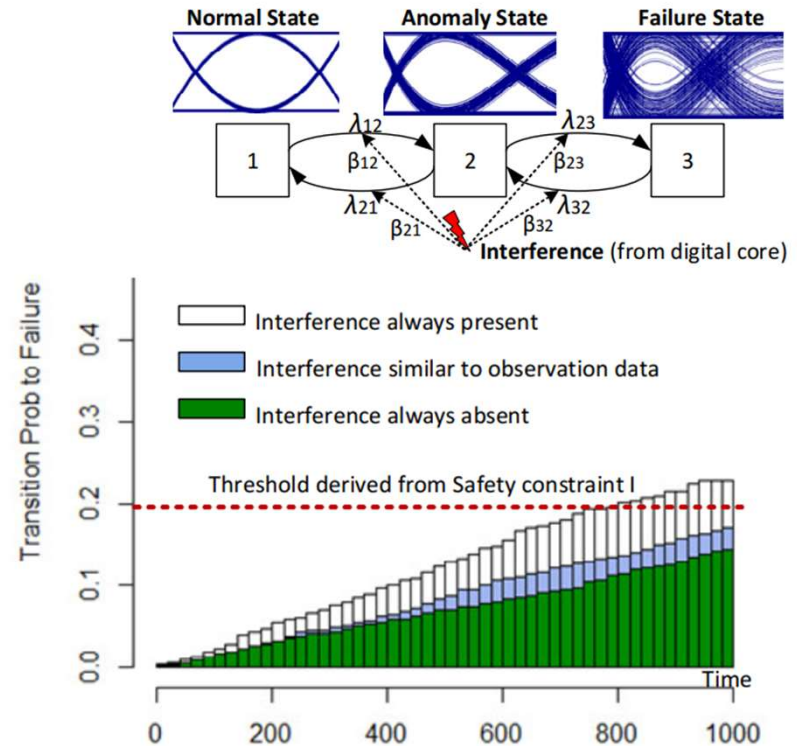
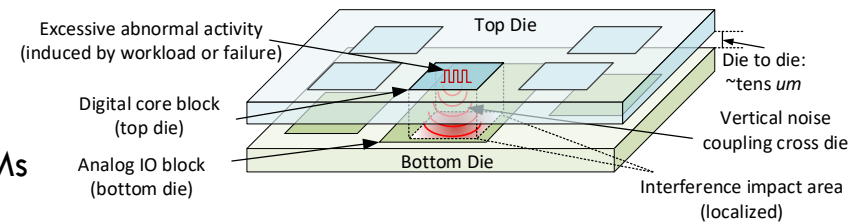
USE CASE EXAMPLE

Failure prediction:

- Predictive maintenance for hyperscalers and automotive OEMs
- Estimate its Remaining Useful Lifetime (RUL)



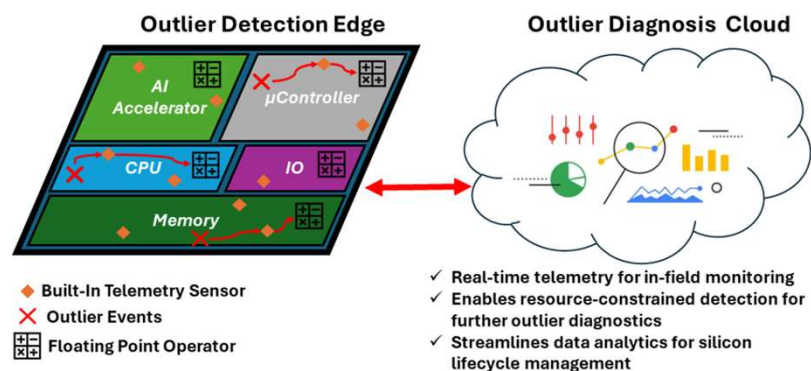
(ITC'20)



USE CASE EXAMPLE

E-SCOUT: Efficient-Spatial Clustering-based Outlier Detection through Telemetry

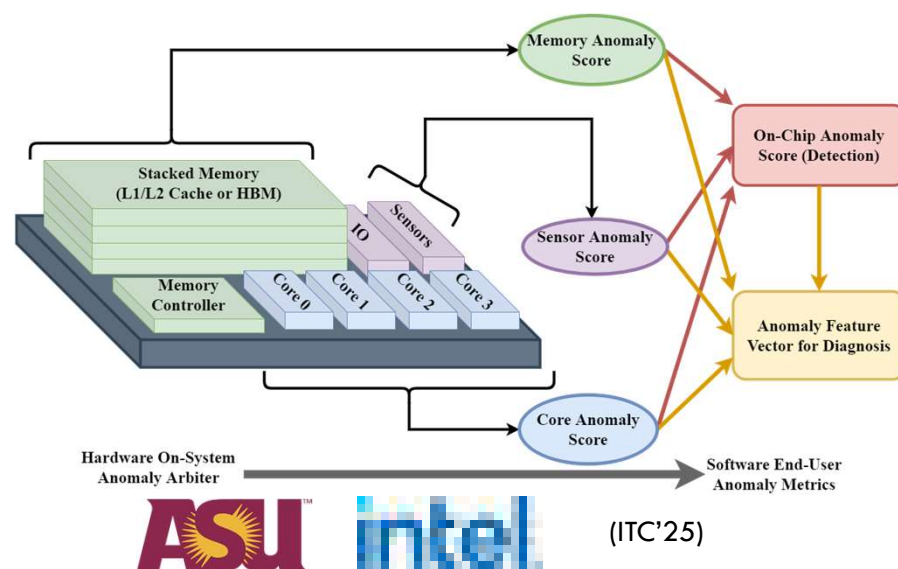
- Online compute- and memory-efficient unsupervised 32-bit floating point outlier detection mechanism
- Telemetry features through performance counters (model-specific register MSR space)



(ITC'24)

On-Chip Telemetry-based Anomaly Notification Engine (OCTANE)

- A two-layer SLM Hardware/Software co-designed framework for low cost
- Provides effective unsupervised on-chip outlier detection performance



(ITC'25)

USE CASE EXAMPLE

(R. Kundu, F. Su, P. Goteti JETTA'19)

Packet dropping at switches
(FC, CSW, RSW)

- Shows up as data loss in servers

Monitor anomalies at source
and destination simultaneously

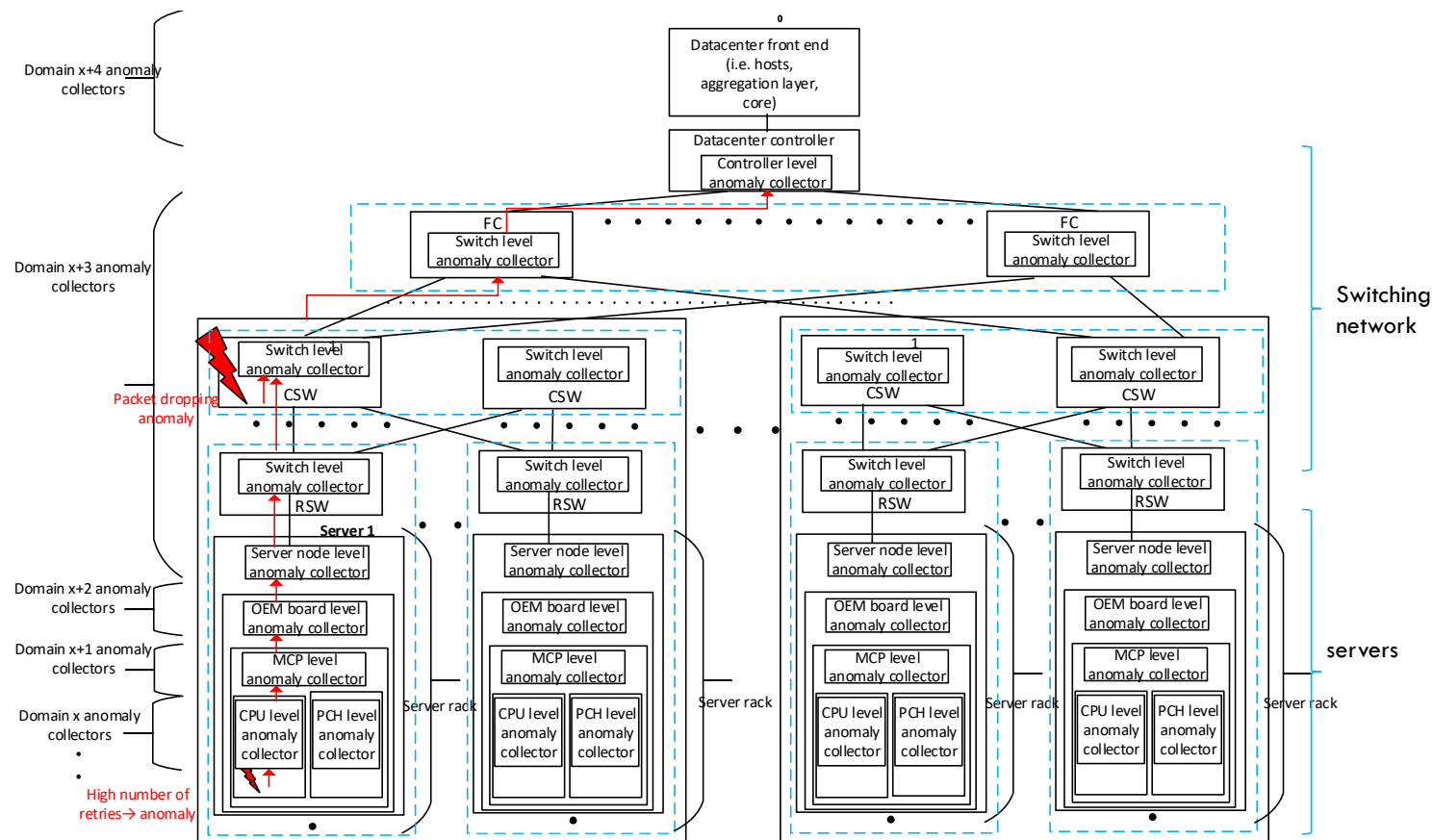
Establish causality between
switch and CPU level anomaly

Route around the anomalous
switch

Avoid datacenter downtime



(JETTA19)



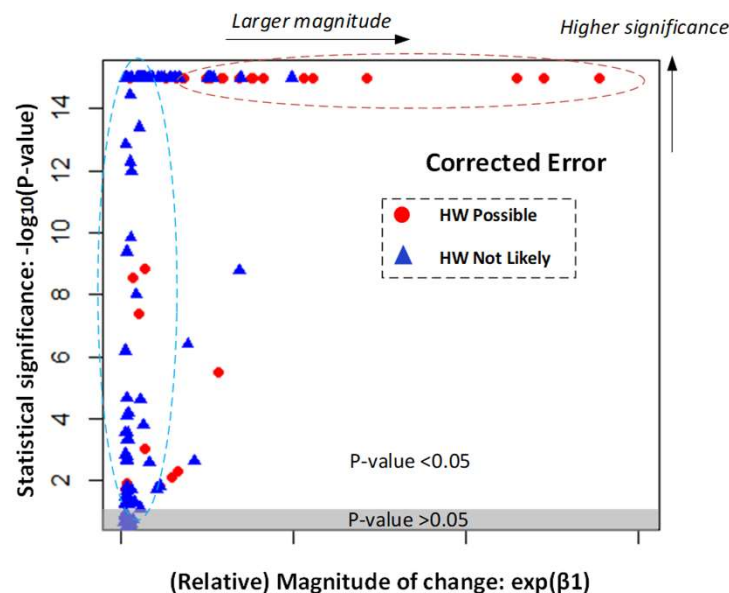
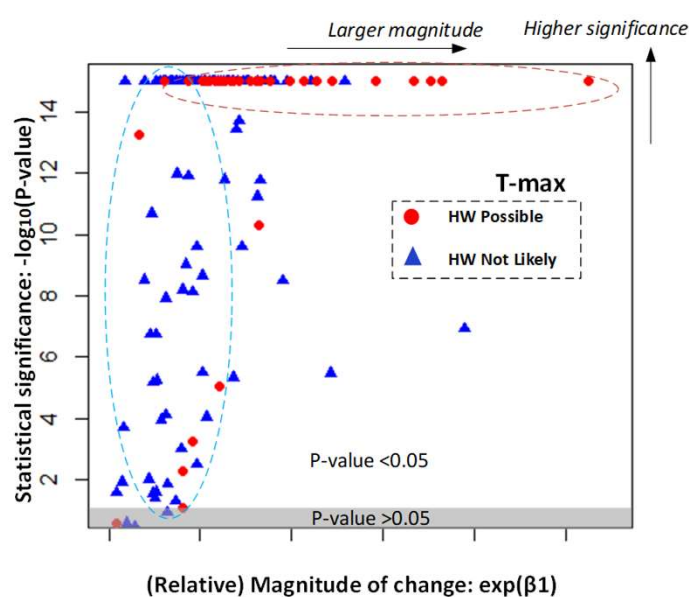
USE CASE EXAMPLE

Telemetry data study: six Intel® SKUs 196,994 PCs over one year (Oct 2021 to Oct 2022)

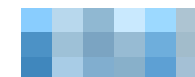
→ 35,011,435 total days of operation

Provide insights to assist RMA: some Uncorrected Error Event is more likely a CPU HW issue or other issue (e.g., SW)?

Measure	Telemetry	Detected by
Temperature	Model Specific Register	CPU
Corrected Error	Windows Event 19	CPU Machine Check Architecture
Bug check (BSOD)	Windows Event 41	Windows OS

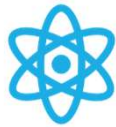


Each datapoint represents one of the 30 studied bug check codes for a CPU SKU



(F. Su, et. al IEEE D&T 24)

CONCLUSION



Growing complexity in the age of heterogeneous integration driving a need for innovation of quality and dependability solutions



Emerging Silicon Lifecycle Management SLM solution based on data-driven telemetry architecture



This presents emerging opportunities to test and DFX community

New **DFT** “Design for Test” → “Design for Telemetry”

New **DFD** “Design for Debug” → “Design for Dependability”

New **DFM** “Design for Manufacturability” → “Design for lifecycle Management”

.....

DFX: “Design for Excellence” or “Design for Everything”

调研内容建议

调研主题：

- 1) **AI赋能可测性设计 (AI for DFT)** 的前沿研究
 - 2) **芯片全生命周期管理 (Silicon Lifecycle Management)** 的前沿研究
- 任选其一即可

阅读文献:不少于 **10** 篇，其中精读不少于**3** 篇

要求：报告正文不少于 **5000**字



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